Artificial Intelligence Review (2026) 59:34
https://doi.org/10.1007/s10462-025-11433-1

Reinforcement learning and the Metaverse: a symbiotic collaboration

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Received: 26 May 2025 / Accepted: 19 October 2025 / Published online: 11 December 2025
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Abstract
The Metaverse is an emerging virtual reality space that merges digital and physical worlds and provides users with immersive, interactive, and persistent virtual environments. The Metaverse leverages multiple technologies, including digital twins, blockchain, artificial intelligence, extended reality, and edge computing to realize the seamless connectivity and interaction between both worlds: physical and virtual. Artificial Intelligence (AI) empowers intelligent decisions in such complex dynamic environments. More specifically, Reinforcement Learning (RL) is uniquely effective in the context of Metaverse applications due to the natural process of learning through interaction and its modeling of sequential decision making, allowing it to be flexible, dynamic, and able to discover complex strategies and emergent behavior in complicated environments where programming explicit rules is impractical. Although multiple works have explored the research on the Metaverse and AI-based applications, there remains a significant gap in the literature that addresses the contribution of RL algorithms within the Metaverse. Therefore, this review presents a comprehensive overview of RL algorithms for Metaverse applications. We examine the architecture of Metaverse networks, the role of RL in enhancing virtual interactions, and the potential for transferring learned behaviors to real-world applications. Furthermore, we categorize the key challenges, opportunities, and research directions associated with deploying RL in the Metaverse.

Keywords Metaverse · Artificial intelligence · Reinforcement learning · Sim2Real · Human feedback

1 Introduction

1.1 Background and motivation

The Metaverse is a collective shared space that converges virtually; seamlessly integrating digital and physical realities while allowing interactions between users in immersive, interconnected virtual spaces (Sadeghi-Niaraki et al. 2025). The concept of the Metaverse has

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evolved significantly from its philosophical origins in Sci-Fi novels and movies to its current technological manifestations (Dwivedi et al. 2022). The early 2000 s, with the launch of Second Life by Linden Lab in 2003, increased the popularity of virtual worlds that enabled the creation of shared 3D virtual spaces for users to explore and interact (Park and Kim 2022). This was followed by the launch of Roblox in 2006, which integrated user-generated content and quickly became a major platform for virtual experiences. Technological advancements continued with Google's 2007 introduction of Street View, followed by the creation of Bitcoin (Dwivedi et al. 2022; Park and Kim 2022). By the 2010 s, the Metaverse concept gained momentum with blockchain technology through Ethereum in 2015, enabling decentralized applications and smart contracts. The emergence of virtual currencies and augmented reality games like Pokémon GO in 2016 further accelerated the popularity of the Metaverse.[1] Given the rising interest and the profitability of these games and worlds as a "proof-of-concept", global giants began to invest in the realization of the Metaverse. Major milestones included Facebook's substantial investment in Metaverse technologies and their 2021 re-branding to Meta as well as Microsoft's introduction of Mesh for virtual collaboration. The industrial Metaverse concept also emerged in 2022 with initiatives such as, Siemens and Nvidia's collaboration, which demonstrated how Digital Twins (DTs) can be integrated into virtual platforms to represent and interact with complex industrial systems. DTs serve as high-fidelity virtual replicas of physical entities, continuously updated via real-time data streams to enable monitoring, analysis, and control (Chen et al. 2024d). While DTs are a foundational component of the Metaverse, the two concepts remain distinct: the Metaverse refers to a persistent, shared, and immersive environment that integrates multiple DTs alongside human avatars and AI agents into a cohesive socio-technical system. In contrast, a virtual environment does not necessarily correspond to real-world objects or systems; instead, it can represent fully synthetic worlds that facilitate experimentation, exploration of new phenomena, or scenarios beyond the constraints of physical reality (Negri and Abdel-Aty 2023; Sai et al. 2024). As such, the Metaverse is increasingly envisioned as the next evolutionary stage of the Internet, enabling individuals and organizations to interact through immersive digital counterparts in a parallel, fully virtualized ecosystem (Wang et al. 2023a).

The Metaverse is the next frontier in digital exploration, where individuals can virtually interact, work, play, and create virtual worlds. Realizing this new paradigm requires the integration and collaboration of enabling technologies such as blockchain (Haber and Stornetta 1991; Nakamoto 2008), artificial intelligence (McCarthy et al. 2006; Fadhel et al. 2024), extended reality (Sutherland et al. 1965; Cardenas-Robledo et al. 2022), and edge computing (Satyanarayanan 2017). Within AI advancements, RL is a machine learning (ML) technique where an agent learns optimal behaviors by performing actions in an environment to maximize rewards. In supervised ML problems, labeled data guides the learning process. However, RL relies on interactions with the environment and feedback in the form of rewards or penalties. Reinforcement learning is more challenging than the traditional ML model because it involves exploration, stability, and reward modeling (Li 2017).

With their recent success, RL-based models have been widely adopted across various domains, such as video games (ElDahshan et al. 2022), robotics (Kober et al. 2013), and autonomous driving (Kiran et al. 2021). Furthermore, they have also been adopted in diverse industrial applications such as Netflix, Amazon, Spotify, etc, to enhance their recommender systems and deliver personalized user recommendations (Kou et al. 2024). Naturally, rein-

[1] https://www.techtarget.com/searchcio/tip/History-of-the-metaverse-explained.

forcement learning has become a key enabling technology for the Metaverse by providing a framework for autonomous agents to interact within dynamic and complex environments of the Metaverse. Figure 1 illustrates the growing research at the intersection of RL and the Metaverse. Reinforcement learning aids in learning from experience and optimizing real-time decisions to adapt to unpredictable and evolving scenarios. By integrating real-world data and feedback into RL algorithms, virtual experiences in the Metaverse can be enriched with real-world context and relevance for immersive and meaningful interactions. Vice versa, the Metaverse provides a highly dynamic and realistic virtual environment for cost-effective reinforcement learning agents to train for future deployment.

1.2 Gaps in the literature

The current body of literature includes numerous surveys that focus on high-level perspectives of the Metaverse, covering its conceptual foundations, system architectures, and evolution. While these works map the ecosystem from technological and security standpoints, they leave unexplored the unique role of RL in shaping adaptive and intelligent Metaverse systems, which is the central focus of our review. Numerous surveys offer comprehensive analyses of the Metaverse's ecosystem, including its underlying technologies, governance structures, and deployment challenges (Wang et al. 2023a; Chen et al. 2024g; Khan et al. 2024). Wang et al. (2023a) provides an overview of the Metaverse, exploring its development status, relevant policies, statistics, and characteristics. Chen et al. (2024g) explores how emerging Metaverse technologies can support and enhance the development of smart cities, reviews current applications and key enabling technologies, and outlines future opportunities and challenges at their intersection to guide researchers and developers. Khan et al. (2024) explores the integration of the Metaverse with wireless systems, its design aspects, key enablers, and high-level architecture, addressing network management, reliability, and security challenges, future directions and solutions for optimizing Metaverse-based wireless

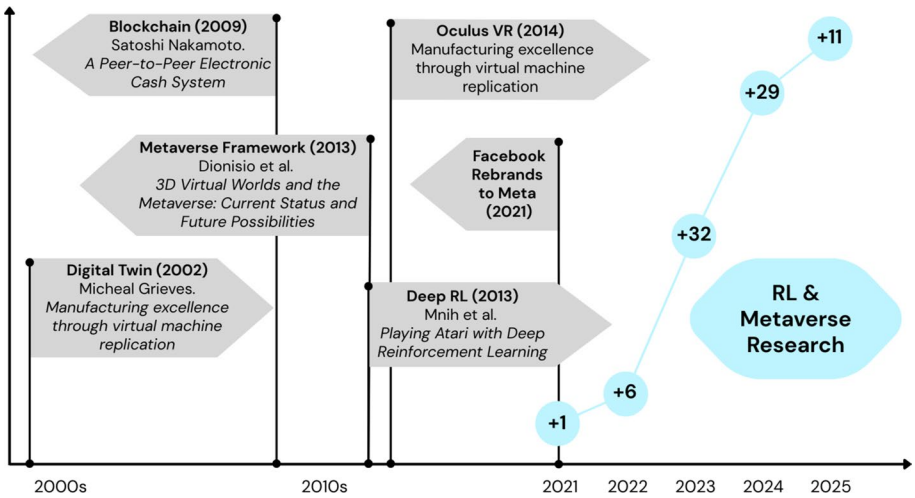

Fig. 1 Key milestones in Metaverse and RL research. The literature at the intersection of RL and the Metaverse considered in this survey is illustrated by the datapoints

applications. These works typically aim to map the current landscape, identify technological enablers, and propose high-level architectural models to guide future development.

As these immersive technologies proliferate, concerns around security and user privacy have become central in the literature with some surveys providing detailed taxonomies of the risks associated with identity, data privacy, and user safety in virtual environments (Wang et al. 2023f; Sharma et al. 2024; Otoum et al. 2024). Wang et al. (2023f) surveys the fundamentals of the Metaverse and includes a novel distributed architecture while addressing security and privacy challenges and countermeasures. Sharma et al. (2024) focuses on the critical need for user safety and security in the Metaverse, highlighting current shortcomings in protecting against threats like harassment, identity theft, and data misuse. It presents a taxonomy of user-centric security issues, reviews existing solutions, and proposes a Zero-Trust-based Metaverse Security Architecture that emphasizes user control, privacy, and proactive protection within immersive virtual environments. Otoum et al. (2024) provides a comprehensive examination of the security and privacy challenges inherent in the development and deployment of the Metaverse. It explores how advanced technologies such as Artificial Intelligence (AI), Extended Reality (XR), 6 G, and blockchain both enable and complicate the security landscape of the Metaverse. The survey critically reviews current state-of-the-art security measures and identifies key open research directions, particularly around cloud-based control systems and the need for scalable, adaptive, and user-centric security frameworks. The authors propose a novel distributed Metaverse architecture to enhance user security and resilience across interconnected 3D worlds. These works also explore architectural and algorithmic solutions, including trust models and distributed systems, designed to mitigate vulnerabilities introduced by the Metaverse's complexity and scale. The emphasis is often placed on designing frameworks that ensure proactive protection while preserving user agency and control.

Several studies have examined the key technologies that make the Metaverse possible, often focusing on the technical capabilities required to deliver immersive, low-latency, and scalable experiences. For instance, Khan et al. (2024), Zhao et al. (2024) and Wang et al. (2024b) explore wireless communication and sensing technologies, highlighting their role in ensuring seamless user interaction. Zhao et al. (2024) explores the integration of intelligent wireless sensing technologies with the Metaverse, focusing on how non-contact, AI-enhanced sensing, such as localization, activity recognition, and pattern analysis, can improve user interaction and access to virtual environments. Wang et al. (2024b) explores the application of enabling technologies like AI, 6 G, Edge Intelligence, and blockchain, and summarizes Integrated Sensing, Communication, and Computing solutions across various scenarios. Edge computing, reviewed in works such as Patra et al. (2024), Xu et al. (2023) and Baidya and Moh (2024), is emphasized for its ability to offload computation and reduce latency. Xu et al. (2023) explores the edge-enabled Metaverse, its architecture, current developments, and key challenges in communication, networking, computation, and blockchain technologies. Patra et al. (2024) focuses on the critical role of edge computing in enabling Metaverse applications by enhancing real-time responsiveness, reducing latency, and supporting immersive experiences. It reviews enabling technologies like 5 G, IoT, and edge AI, examines real-world case studies, addresses key challenges and solutions, and outlines future research directions to guide the continued development of edge-enabled Metaverse ecosystems. Baidya and Moh (2024) focuses on resource allocation in edge-enabled Metaverse environments, emphasizing its critical role in supporting immersive, low-latency, and

scalable virtual experiences. Other contributions investigate the integration of AI (Huynh-The et al. 2023) and soft computing (Tavana and Sorooshian 2024), while Zhu et al. (2024) delves into emerging human–machine interfaces such as brain-computer interfaces (BCIs) with AI modeling of human emotion, cognition, thoughts, and body movements.

Several recent studies have explored AI as an emerging technology relevant to the development of the Metaverse (Soliman et al. 2024; Huynh-The et al. 2023; Otoum et al. 2024; Wang et al. 2024b; Patra et al. 2024). Soliman et al. (2024) explores the role and contributions of AI in the Metaverse and considering user privacy, bias, and discrimination issues. Huynh-The et al. (2023) investigates the role of AI in enhancing immersive experiences and virtual interactions, while Otoum et al. (2024) focuses on how machine learning can address security and privacy vulnerabilities in the Metaverse. In a related domain, Hu et al. (2024b) reviews methods to bridge the "reality gap" in autonomous driving through techniques such as sim-to-real transfer, digital twins, and parallel intelligence; highlighting the importance of simulation in AI-based systems. Although AI has received attention in the context of the Metaverse, the literature largely falls short in addressing the following critical aspects:

- Many surveys focus on the high-level frameworks of the Metaverse (Tavana and Sorooshian 2024; Wang et al. 2023a, f) or narrow application areas such as wireless technologies (Khan et al. 2024) or brain-computer interaction (Zhu et al. 2024). However, none of these synthesizes insights related to reinforcement learning (RL) in a Metaverse context.
- Although reinforcement learning is a foundational technique for building adaptive and intelligent agents, particularly in dynamic and interactive environments, it is often only briefly mentioned as a subset of broader AI discussions (Soliman et al. 2024; Huynh-The et al. 2023; Otoum et al. 2024; Wang et al. 2024b; Patra et al. 2024) or in specific Metaverse applications (Baidya and Moh 2024). The existing literature does not discuss the specific challenges, architectures, or opportunities associated with the deployment of RL within the Metaverse.
- Despite the growing relevance of simulated environments in the training of RL agents, especially in simulation-to-reality (Sim2Real) contexts (Hu et al. 2024b), most of the literature overlooks the role of simulation in Metaverse research. This gap limits researchers seeking appropriate tools for prototyping and testing intelligent systems in Metaverse settings.
- Human-in-the-loop reinforcement learning, where expert feedback guides agent behavior, has shown promise in other domains but remains unaddressed in Metaverse-related RL studies. This feedback mechanism is especially valuable in immersive environments, where safety, adaptability, and personalization are critical.

Table 1 summarizes the key differences in the literature compared to this work. To the best of our knowledge, no prior survey or review offers a dedicated analysis of reinforcement learning applications in the Metaverse.

1.3 Scope and objective

This study aims to bridge a significant gap in the literature by providing a comprehensive overview of reinforcement learning applications in the Metaverse. It emphasizes the critical

Table 1 Existing surveys for the Metaverse

References	Metaverse	Key Focus	RL	Human feedback	Sim2Real	Simulators
Wang et al. (2023a)	✓	Development and Policies	✗	✗	✗	✗
Wang et al. (2023f)	✓	Architecture and Security	✗	✗	✗	✗
Chen et al. (2024g)	✓	Smart Cities	✗	✗	✗	✗
Sharma et al. (2024)	✓	Security and Privacy	✗	✗	✗	✗
Otoum et al. (2024)	✓	Machine Learning for Security	✗	✗	✗	✗
Wang et al. (2024b)	✓	Integrated Sensing, Communication, and Computing	✗	✗	✗	✗
Huynh-The et al. (2023)	✓	Artificial Intelligence	✗	✗	✗	✗
Zhu et al. (2024)	✓	Brain-Computer Interface	✗	✗	✗	✗
Patra et al. (2024)	✓	Edge Computing	✗	✗	✗	✗
Zhao et al. (2024)	✓	Wireless Sensing	✗	✗	✗	✗
Tavana and Sorooshian (2024)	✓	Soft Computing	✗	✗	✗	✗
Xu et al. (2023)	✓	Edge Computing	✗	✗	✗	∼
Baidya and Moh (2024)	✓	Edge Computing (Resource Allocation)	∼	✗	✗	✗
Soliman et al. (2024)	✓	Artificial Intelligence	∼	✗	✗	✗
Khan et al. (2024)	✓	Wireless Systems	✗	✗	✓	✗
Hu et al. (2024b)	✗	Autonomous Driving	∼	✗	✓	✓
This work	✓	Reinforcement Learning	✓	✓	✓	✓

Partially included

roles of simulation tools, Sim2Real transfer, and human feedback in shaping this emerging research area. This work provides the first review to explore both how RL can enhance Metaverse experiences and how the Metaverse itself can serve as a testbed for real-world RL challenges.

This work provides a detailed discussion of simulation platforms tailored for RL-driven Metaverse applications; an area that remains underexplored in existing literature. The review also introduces the use of human feedback as a key mechanism for creating adaptive, safe, and intelligent agents within immersive environments.

In addition to surveying current developments, this review analyzes the challenges and identifies potential research directions for expanding the RL-based Metaverse ecosystem. By integrating these dimensions, the study presents a holistic view of the state of RL in the context of Metaverse technologies.

Focusing specifically on literature at the intersection of RL and the Metaverse, the review systematically categorizes existing work based on how RL contributes to different components of Metaverse architecture. It offers insights into how RL algorithms are being adapted and optimized to tackle the unique challenges posed by distributed virtual environments. The exploration is guided by two main research questions:

- **RQ1:** *How has reinforcement learning been integrated into various modules of the Metaverse? How can different RL algorithms enhance its components, processes, and services?*
- **RQ2:** *How can the Metaverse serve as a platform to improve the performance and applicability of RL algorithms in real-world contexts?*

To address these questions, the literature is examined in terms of the RL algorithms used, the roles of RL agents, and the types of problems they are designed to solve. Special attention is given to the complexity and high-dimensional nature of Metaverse environments, emphasizing the value of simulation platforms that offer researchers detailed data access and high degrees of control and customization within virtual environments. Furthermore, our review analyzes how RL algorithms, agent roles, and problem settings intersect with Metaverse requirements. We emphasize on the role of simulation platforms, with their fine-grained control and human-in-the-loop integration, and the broader ways in which RL emerges as a core enabler of adaptive, interactive, and transferable intelligence within immersive virtual environments.

1.4 Paper structure

The remainder of the paper is outlined as follows. Section 2 provides preliminary details and definitions regarding the Metaverse and reinforcement learning. Section 2.1 focuses on the Metaverse, its general architecture, its enabling technologies, and state-of-the-art industrial efforts. Section 2.2 outlines the reinforcement learning paradigm and the taxonomy of reinforcement learning algorithms in the literature. Section 3 explores the role of reinforcement learning in supporting the various requirements of the Metaverse within its modules. Section 4 explores how the Metaverse can be used to build reinforcement learning models for real-life decisions, highlighting the existing simulation tools that are compatible with RL and, more specifically, human-in-the-loop implementations. The challenges and future

directions in this emerging field are discussed in Sect. 5 and Sect. 6, respectively. Finally, Sect. 7 summarizes the work, highlighting the key contributions and conclusions.

2 Preliminaries

In preparation for the discussions and analyses in this review, key concepts relating to the Metaverse and reinforcement learning are defined in the following sections.

2.1 Metaverse and related technologies

The term *metaverse* is a portmanteau of the prefix *meta*, meaning of a higher-order, and *universe*; first coined in the 1992 Sci-Fi novel, Snow Crash. *Metaverse* described a parallel virtual world accessible through digital avatars and virtual reality glasses (Sadeghi-Niaraki et al. 2025). Before this, William Gibson's 1984 novel Neuromancer introduced the notion of cyberspace, a concept later popularized in the 1999 film *The Matrix*, which depicted a dystopian reality controlled by intelligent machines.[2] [3] The term is used, holistically, to describe the gradual convergence of our physical world with the digital one (Huynh-The et al. 2023). Presently, it is used to describe one or more persistent, shared, and linked 3D virtual spaces in which users can interact and collaborate in an immersive experience (Huynh-The et al. 2023). This study uses "*the Metaverse*" to refer to a modern class of systems containing a set of connected virtual 3D spaces, with virtual representations of physical entities and users, facilitating an immersive experience, ownership, interaction, and synchronicity with the real-world.

2.1.1 High level architecture of the Metaverse

The architecture of the Metaverse, illustrated in Fig. 2 can be represented in three main modules: virtual space, physical space, and connectors or enablers to link the virtual space with the physical space. The design requirements of these spaces are such that they (Wang et al. 2023a; Abilkaiyrkyzy et al. 2023):

- **Guarantee**: Interoperability, Security, Persistence, and Accessibility,
- **Maintain**: Connectivity, Immersiveness & Realism, Scalability, Configurablity, and Privacy & Ethics,
- **Support**: Social Interaction, Multi-modal Interaction, Decentralization, Artificial Intelligence, and Market Access.

Virtual Consists of the 3D virtual environment where users can interact with fully synthetic spaces. It can host digital twins (digital replicas of physical entities) and provides a blend of real-world data and simulated content to enable immersive experiences. Furthermore, real-world activities can be replicated within the virtual environment and extended beyond the limitations of the physical world.

[2] https://medium.com/@angietaylor197411/the-metaverse-is-it-the-matrix-unfolding-c42f636ccea1.

[3] https://www.inverse.com/entertainment/star-wars-skeleton-crew-trailer-d23.

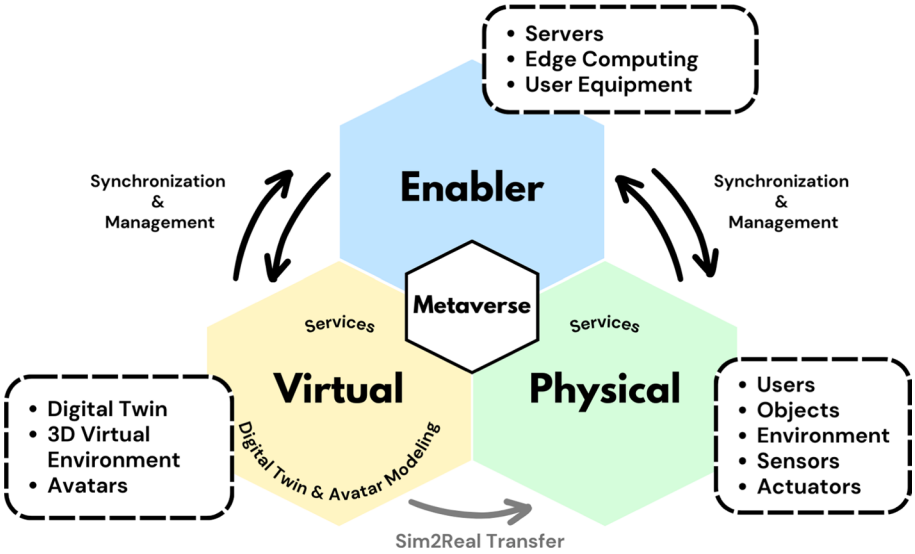

Fig. 2 The modules of Metaverse architecture and their interactions

It accurately models, simulates, and augments real-world entities and their interactions for immersive experiences. Furthermore, real-world activities can be replicated within the virtual environment and extended beyond the limitations of the physical world. Users hold ownership over their virtual replicas, *avatars*, and use them to interact with other virtual entities. The virtual world in the Metaverse can provide services to users, acting as both a social space and an open market. For example, virtual musical concert attendance provides users with an immersive concert experience while transcending the capacity limitations of physical venues and allowing attendees to join from anywhere in the world (Jin et al. 2022). Another use-case allows the Metaverse to be used as a complex multi-actor simulation playground for *Sim2Real Transfer* (Zhao et al. 2020). For example, training autonomous vehicles in the Metaverse under multiple traffic scenarios and weather conditions in various situations without real-world risks before deploying into real-world vehicles (Salvato et al. 2021). Physical The real world that contains objects, environments, and interactions. The physical space is the foundation of the digital twin to replicate real-world data and scenarios in the Metaverse. Realizing an immersive virtual world and Metaverse experience requires an abundance of sensors and actuators in the physical space. Sensors capture the state of the real world and actuators respond and reflect virtual changes back into the physical space. Enablers The physical devices that facilitate the connection between the virtual and physical spaces. They represent seamless communication and interaction and include network infrastructures, user equipment, and intelligent algorithms that aid in the interaction and the flow of information between the real world and the Metaverse. The enablers work to synchronize the virtual and physical spaces and maintain a high-quality continuous connection through efficient scheduling and resource management. In order to facilitate seamless interaction between the virtual and physical worlds, the activities of the enablers can be categorized as *resource management* and *synchronization*. Within resource management, intelligent designs are needed to efficiently make use of computing resources, wireless resources, and

the mobility of the enablers. Synchronization activities consider efficient data collection, quantization, sampling rates, and packet error minimization, in addition to the underlying resources' management.

2.1.2 Technologies for the Metaverse

The Metaverse leverages multiple emerging technologies (Pervez et al. 2024) to create its immersive digital landscape as illustrated in Fig. 3. For instance, digital twins (DTs) provide virtual replicas of physical entities like cities or buildings in the Metaverse. Augmented reality (AR) and virtual reality (VR) are crucial in shaping the user experience in the Metaverse. Similarly, blockchain technology ensures security, transparency, and decentralized control within the Metaverse. Artificial intelligence (AI) powers advanced functionalities within the Metaverse, such as natural language processing for interactive virtual assistants, personalized content recommendations, and adaptive simulations that respond to user actions in real-time (Soliman et al. 2024).

Digital twins Digital Twins (DTs) are virtual representations of objects or systems specifically designed to accurately reflect the complexity of the physical objects and their components. DTs are well-suited for emulating real-life systems/environments since they are capable of modeling the life cycles of their physical counterparts, in addition to their flexibility in being updated using real-time data. Moreover, DTs incorporate intelligent decision-making by employing simulation, reasoning, and machine learning techniques. These digital twins allow for real-time simulation and analysis, enabling efficient management and optimization of physical assets, and they can be integrated into the virtual worlds of the Metaverse for visualization and interaction.

Augmented reality (AR) and virtual reality (VR) 3D reconstruction creates a detailed three-dimensional model from real-world objects or environments. It is crucial for building realistic virtual spaces and objects, enhancing user immersion and interaction capabilities

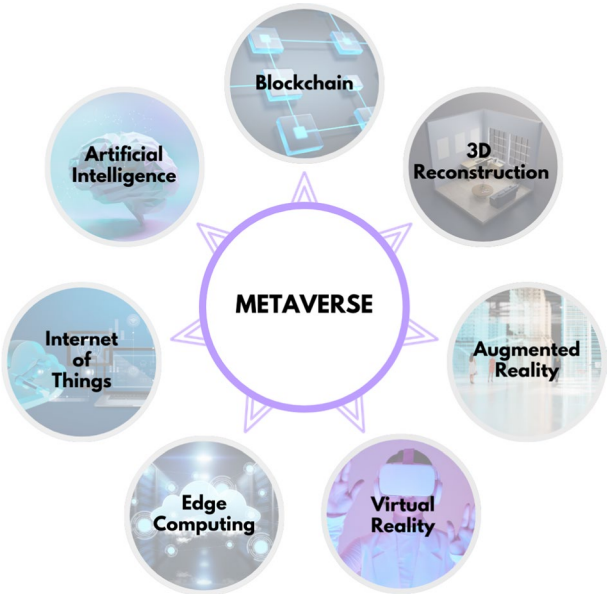

Fig. 3 The technologies utilized to develop the Metaverse

within the digital environment. AR utilizes digital content in the real-world environment, enriching perception and interaction with physical surroundings. AR enhances interaction by blending digital and physical worlds, allowing users to engage with digital elements superimposed on their environment. While VR allows users to interact with each other and digital elements in a virtual environment, it provides an entirely virtual experience in a life-like setting. Unlike AR, which blends digital content with the real world, VR immerses the user completely into a virtual space, often through the use of specialized headsets. It enables interaction not just with the environment but also with other users within the virtual space.

Blockchain It facilitates secure transactions, ownership verification of virtual assets, and transparent transactions among the users. Moreover, it offers a decentralized framework for the Metaverse, making it more equitable and interoperable.

Edge computing It optimizes data processing and reduces latency by decentralizing computation closer to the user or device, enhancing the responsiveness and performance of applications and services within the Metaverse.

Artificial intelligence Data-driven approaches enable digital systems to detect, comprehend, act, and learn by simulating human intelligence. These models can provide autonomous solutions in Metaverse systems, which enhances the hardware and computation utilization as well as the user experience through machine learning, deep learning, neural networks, natural language processing, object detection, and pattern recognition.

2.2 Reinforcement learning

Reinforcement learning (RL) is an innovative field in AI that draws from the learning behavior of animals through simple reinforcement mechanisms originating in the studies of psychologist B. F. Skinner in the 1930 s.

RL is an interdisciplinary subset of optimal control theory and machine learning, in which an *agent* interacts with the world and periodically learns in its dynamic environment by receiving rewards based on the actions it takes. The agent navigates a balance between the *exploration* of uncharted territory and the *exploitation* of accumulated knowledge while maximizing cumulative rewards reflecting the agent's ability to achieve its goal (Arulkumaran et al. 2017).

In the RL paradigm, Markov Decision Processes (MDPs) or, more generally, Partially Observable Markov Decision Processes (POMDPs), are the underlying stochastic models for decision making. The MDP can be described by the tuple (S, A, P, R, γ), where S represents the *state space* of the agent, and A is the set of *actions* that the agent can take in the environment. $P : S \times A \times S \to [0, 1]$ denotes the *transition probability*, modeling the dynamics of state transitions based on the actions. $R : S \times A \times S \to \mathbb{R}$ denotes the reward recieved by the agent after taking action $a \in A$ at a given state $s \in S$ for which there is a maximum value $R_{\max}$ and $\gamma \in (0, 1)$ is the *discount factor* that determines the importance of future rewards compared to the immediate ones. The RL agent navigates an environment in a sequence of time steps $t \in \{1, 2, 3, \ldots\}$; at each time t, the agent performs an action $a_t \in A$, transitions from one state s_t to another s_{t+1}, and collects a reward r_t. The agent chooses the action based on a *policy* function $\pi \in \Pi$, which may be *deterministic*, specifying a unique action a for each state s, or *stochastic*, whereby actions are selected according to a probability distribution function.

The goal of the RL agent is to maximize the cumulative (expected) discounted *rewards* defined by the *value* function $V^{\pi}(s) : S \to \mathbb{R}$ defined for all t by Eq. 1. As a result of (Puterman 2014), there exists an optimal deterministic policy π^* which uniquely maximizes the

value function. Thus, in order to solve the optimization problem given by Eq. 2, most RL algorithms strategically choose deterministic policies.

$$V^{\pi}(s) = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R(s_{t+k}, a_{t+k}, s_{t+k+1}) \mid s_t = s, \pi\right], \forall t \tag{1}$$

$$V^*(s) = \max_{\pi \in \Pi} V^{\pi}(s) \tag{2}$$

The recursive behavior of the value function follows the Bellman equations defined by Eqs. 3 and 4.

$$V^{\pi}(s) = \sum_{s' \in S} P(s, \pi(s), s') \left[R(s, \pi(s), s') + \gamma V^{\pi}(s')\right], \quad \forall \pi \in \Pi \tag{3}$$

$$V^*(s) = \max_{a \in A} \left[r(s, a) + \gamma \sum_{s' \in S} P(s, a, s') V^*(s')\right], \quad \forall s \in S \tag{4}$$

where for every state-action pair (s, a),

$$r(s, a) = \mathbb{E}_{s' \sim P(s,a,\cdot)} R(s, a, s') \tag{5}$$

Another function of interest is the Q-value function $Q^{\pi}(s, a) : S \times A \to \mathbb{R}$ defined in Eq. 6, which is the expected total discounted reward following policy π given an initial state s and initial action a. Similarly, the Q-value function satisfies the Bellman equation as indicated by Eq. 7 and the optimal Q-value functions for each state-action pair (s, a), are given by Eq. 8 which allow the direct computation of the optimal policy by using Eq. 9.

$$Q^{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) \mid s_0 = s, a_0 = a\right] \tag{6}$$

$$Q^{\pi}(s, a) = \sum_{s' \in S} P(s, a, s') \left[R(s, a, s') + \gamma Q^{\pi}(s', \pi(s'))\right] \tag{7}$$

$$Q^*(s, a) = \max_{\pi \in \Pi} Q^{\pi}(s, a) \tag{8}$$

$$\pi^*(s) \in \arg \max_{a \in A} Q^*(s, a) \tag{9}$$

2.2.1 Reinforcement learning algorithms

This subsection provides a short summary and taxonomy of reinforcement learning (RL) algorithms. RL algorithms can be distinguished by the aspects of the underlying MDP they represent and the approaches used to update the representations during the learning phase (Ribeiro 2002).

In RL, the terms *model-free* and *model-based* are used to indicate whether, during training or testing, the agent utilizes a prediction of the change in the environments. Model-based methods are difficult to implement for higher-dimensional state spaces or MDPs where the transition probabilities and reward functions are unknown or computationally challenging. For these reasons, RL research in Metaverse systems uses state-of-the-art RL algorithms that aim to balance the amount of state exploration and the exploitation of previous knowledge to approximate optimal solutions.

Model-based RL algorithms In model-based RL algorithms, the agent relies on an explicit model of the environment comprised of the transition dynamics and reward function. Thus, the RL agent learns to predict the reward of its actions' outcomes and greedily chooses the action that maximizes its expected reward. If S and A are finite and discrete, and $\gamma < 1$, then dynamic programming using Value Iteration, Q-Value Iteration, or Policy Iteration algorithms may be used to solve the MDP. Such RL agents are well-suited for scenarios where there is full knowledge of the environment and the effect of all the actions within the environment. Model-based RL algorithms are often used for static or fixed environments where agents can effectively learn and formulate future actions (Polydoros and Nalpantidis 2017).

Model-free RL algorithms In model-free RL algorithms, the agent builds its learning experience by performing actions and observing their rewards without constructing an explicit model of the environment dynamics. These algorithms may sample from their experience memory but do not generate predictions of the next state or next reward. They use the accumulated experience to learn a policy or strategy function, which decides actions aimed at optimizing rewards. Monte Carlo methods can be used to estimate $V^{\pi}(s)$ and $Q^{\pi}(s, a)$ by iteratively sampling actions and the consequent rewards, however, they would require a computationally unrealistic amounts of data. Therefore, other RL algorithms such as SARSA, Q-learning, and Actor-Critic methods are favored for having higher computational efficiency given that they experience sufficient data. The taxonomy of modern model-free RL algorithms is illustrated in Fig. 4, and related concepts and definitions are presented below.

Exploration–exploitation dilemma A central challenge in RL is the exploration–exploitation dilemma, which refers to the trade-off between choosing actions that yield high expected rewards based on current knowledge (*exploitation*) and selecting actions that may lead to new knowledge and potentially greater long-term rewards (*exploration*). Exploitation focuses on leveraging previously visited states to maximize immediate rewards by selecting likely sub-optimal actions. However, this strategy risks premature convergence to suboptimal policies if the agent fails to adequately explore the environment. Alternatively, exploration involves selecting actions with uncertain outcomes, which may initially result in lower rewards but can help uncover better policies in the long run. In order to circumvent such dilemma, RL agents use methods such as ϵ -greedy, soft max action selection, Thompson sampling, and Upper Confidence Bound (UCB), each offering different mechanisms to trade off short-term gains for long-term performance. For example, *Q-learning* uses a parameter ϵ to determine the probability of selecting a random action or greedily choosing the action based on the current best estimate of the Q-value function.

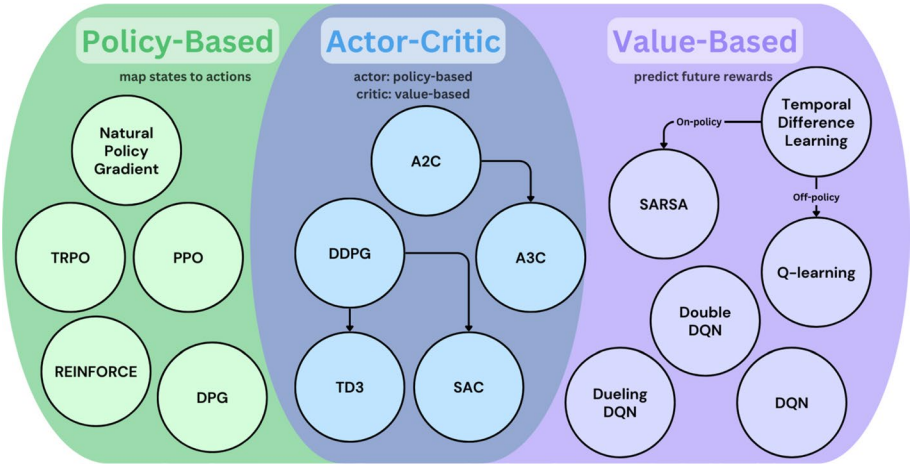

Fig. 4 The taxonomy of modern model-free reinforcement learning algorithms

On-policy vs. off-policy The terms *on-policy* and *off-policy* are used to describe the strategy that an RL algorithm uses to select actions as it is learning. On-policy algorithms, such as SARSA, learn the value of the policy currently being followed by the agent. They update the policy based on the actual actions taken during exploration, which can result in more stable but less efficient learning. Whereas, Off-policy algorithms, such as Q-learning, learn the value of an optimal policy independently of the actions used to collect data. This allows for greater flexibility and sample reuse, often improving learning efficiency. In policy-gradient methods, this distinction can be blurred, especially when both behavior and target policies are stochastic and parameterized similarly.

Policy-based RL algorithms Policy-based RL methods learn a parameterized policy π directly, optimizing it to maximize expected cumulative rewards. These methods are especially useful in environments with high-dimensional or continuous action spaces. Some examples of policy-based RL algorithms include traditional Policy Gradient and its faster-converging counterparts, Natural Policy Gradient, REINFORCE, Trust Region Policy Optimization (TRPO), Deterministic Policy Gradient (DPG), and Proximal Policy Optimization (PPO).

Value-based RL algorithms Value-based methods in RL approximate optimal value functions of the MDP without explicitly storing a policy function. As noted earlier, the policy function is implicit and can be derived directly from the value function by picking the action with the best value. Some examples of value-based RL algorithms include Temporal-Difference Learning (Q-learning and SARSA), Deep Q-Networks (DQN), Double Deep Q-Networks (Double DQN), and Dueling Deep Q-Networks (Dueling DQN). These methods are effective in discrete action spaces and can leverage experience replay for sample efficiency.

Actor-critic RL algorithms Actor-critic architectures for RL merge both value-based and policy-based methods. The algorithms learn two models: 1) the *actor* agent that learns the policy and 2) the *critic* agent that learns to evaluate the actor and is then used to improve the policy. There are several types of Actor-Critic algorithms, including Advantage Actor-Critic (A2C), Asynchronous Advantage Actor-Critic (A3C), Deep Deterministic Policy Gradient (DDPG), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Soft Actor-Critic (SAC). These

hybrid architectures are particularly powerful in continuous action domains and benefit from both stability and flexibility.

Deep reinforcement learning Deep reinforcement learning (DRL) integrates RL with deep learning to enable agents to make decisions in complex, high-dimensional environments (Wong et al. 2023). Traditional RL methods, such as tabular Q-learning, work well in environments with small and discrete state spaces. However, in real-world applications such as, robotic control, game playing, or autonomous driving, the state space can be enormous or continuous, making tabular methods infeasible. DRL uses neural networks as function approximators to estimate value functions or policies, allowing the agent to handle high-dimensional inputs and generalize across similar states. Deep Q-Network (DQN) is a classic example of DRL which uses a convolutional neural network to approximate the Q-value function.

Quantum reinforcement learning Quantum Reinforcement Learning (QRL) enhances classical RL by leveraging principles of quantum computing, such as superposition, entanglement, and quantum interference, to improve learning efficiency, scalability, and decision-making performance. QRL models, adopt hybrid quantum-classical architectures, in which variational quantum circuits (VQCs) serve as function approximators for policies or value functions (Seetohul et al. 2024). VQCs are parameterized quantum circuits composed of quantum rotation gates (Rx, Ry, Rz) and entanglement gates (CNOT or CZ) that connect qubits. These circuits are trained similarly to neural networks by adjusting the parameters that dictate the rotation angles using gradient-based optimization techniques. The entanglement between qubits allows the quantum model to capture complex correlations between input features that may be difficult to learn with classical models. These quantum models leverage phenomena like superposition, entanglement, and quantum interference to enable richer representations and more efficient exploration of the state-action space. Quantum models, which operate on probability amplitudes, are naturally suited for capturing features with inherent uncertainty. QRL also enables more efficient exploration of the action space by leveraging quantum superposition, allowing agents to evaluate multiple strategies simultaneously. Moreover, quantum entanglement allows QRL-based systems to uncover and exploit complex correlations between parameters that classical models may struggle to represent. QRL also facilitates scalable learning across multiple agents and environments by supporting parallel policy updates and coordinated behavior through quantum-enhanced communication.

Mapping of metaverse tasks to RL algorithms While the taxonomy above summarizes key RL families, its practical relevance emerges when mapped to the unique problem structures of the Metaverse. Recent surveys highlight that RL is particularly suited for environments characterized by partial observability, non-stationarity, and real-time adaptation under uncertainty (Nguyen et al. 2020; Zhang et al. 2021; Huh and Mohapatra 2023). For instance, actor–critic and recurrent RL methods have proven effective in managing partially observable multi-user spaces, especially when each user has a different view of the environment without a complete information of all its surroundings (Nguyen and Lin 2023), while multi-agent RL frameworks such as centralized training with decentralized execution (CTDE) explicitly address the non-stationarity that arises in multi-agent interactions (Foerster et al. 2018; Huh and Mohapatra 2023). Constrained and safe RL approaches provide mechanisms to incorporate safety and ethical requirements into policy learning, which is critical in immersive socio-technical systems where misaligned behaviors could have harmful consequences (García and Fernández 2019; Sukhija et al. 2024). Similarly, preference-based RL and reinforcement learning from human feedback (RLHF) enable user-centric adaptation by integrating implicit and explicit human signals into

decision-making (Christiano et al. 2017; Lee et al. 2021). On the systems side, model-based and hybrid RL algorithms are particularly valuable in resource-constrained environments, offering efficient strategies for latency- and energy-aware optimization in edge-assisted Metaverse services (Nguyen et al. 2020; Yu et al. 2022). Finally, deep RL methods combined with digital twins and domain randomization are increasingly used to bridge sim-to-real transfer, allowing policies to be trained and stress-tested in virtual replicas before deployment in live Metaverse environments (Wang et al. 2023d; Hayla et al. 2025). This problem-driven view links the algorithmic taxonomy to the system requirements outlined in Sect. 2.1.1 positioning RL not only as a definitional framework but as a practical toolkit for addressing Metaverse-specific challenges (Table 2).

Use-Case Box 1: Industrial Metaverse Applications

Overview: The Industrial Metaverse transforms how industries operate by integrating digital twins, simulations, and immersive technologies to optimize operations, improve model training, and drive innovation without physical constraints. Global industries and leading technology platforms are shaping this vision.

Nvidia – Omniverse Platform is a key enabler of the industrial Metaverse, providing the foundational technologies through GPUs and AI platforms. Its Omniverse platform allows engineers and designers to build and interact with digital twins and virtual simulations, supporting domains from automotive design to manufacturing. AI-driven predictive maintenance and process optimization further highlight its impact.

Qatar Airways – QVerse is an immersive VR environment for virtual tours of airport facilities and aircraft interiors. With a MetaHuman cabin crew avatar ("Sama"), users can explore premium check-in areas, cabins, and service offerings in an interactive and personalized way, enhancing customer engagement and experience.

Nissan – Invisible-to-Visible (I2V) merges AR and VR to enhance driving by displaying hidden hazards, real-time traffic, and parking availability. Core components include Omni-sensing, Seamless Autonomous Mobility, and ProPILOT. Beyond safety and efficiency, I2V integrates a social layer, enabling passengers to interact with others in the Metaverse through 3D avatars.

Meta – Horizon and Presence Platforms connects Metaverse of social, educational, and professional spaces. Its Horizon suite (Worlds, Workrooms, Venues) supports collaboration and entertainment, while the Presence platform provides AI-powered tools like Spatial Anchors and Scene Understanding for realistic mixed reality. Applications extend to immersive learning, remote work, and virtual business ecosystems.

Takeaway: These cases illustrate how diverse industries leverage digital twins, immersive interfaces, and AI to bring the Industrial Metaverse to life.

Table 2 Problem-driven taxonomy: mapping Metaverse task features to RL families and Sect. 2.1.1 system requirements

Metaverse Task Feature	Challenges	Recommended RL Families	Mapping to Sect. 2.1.1 Requirements
Partial observability (local sensing, occlusion, intermittent telemetry)	Hidden-state inference; needs memory or environment models	Recurrent actor–critic (RNN + A2C/PPO), POMDP-model-based RL, belief-state planners	Perception and robustness (state estimation, graceful degradation)
Multi-agent non-stationarity (co-evolving policies in shared spaces)	Policies change as other agents/users learn → instability	MARL (CTDE like MADDPG, QMIX), opponent modelling, population-based training	Scalability and stability (robust coordination, continual learning)
Safety, constraints, alignment (prevent harmful or unsafe behaviors)	Must guarantee constraints during exploration and deployment	Constrained RL (Lagrangian/CRPO), safe exploration, shielded controllers, reward shaping	Safety, regulatory and ethical requirements
Human-in-the-loop / preference feedback (users shape objectives)	Aligning agent behaviour with human values/preferences	Preference-based RL, RLHF, imitation learning, interactive reward modelling	Usability and alignment (explainability, human oversight)
High-dimensional / complex sim-to-real (rich visuals, physics)	Sample inefficiency, reality gap	Deep RL + representation learning, hierarchical RL, sim-to-real via domain randomization, DT-augmented training	Fidelity, transfer, and deployment (sim-to-real robustness)
Resource & communication constraints (bandwidth, latency, battery)	Limited shared information, intermittent communications	Decentralized policies, communication-aware MARL, CTDE with compressed state	Energy, latency and resilience (operational constraints)

3 Reinforcement learning for the Metaverse

The Metaverse encapsulates a large interconnected network of virtual environments where users interact, socialize, work, and play in a simulated digital twin. Such virtual spaces are immersive in nature and expansive in scope and are built using advanced spatial computing technologies and AR frameworks. Users can navigate diverse landscapes within the Metaverse, including realistic urban simulations, and can benefit from experiences and services such as AI-content generation, extended reality, and gaming. The Metaverse offers various key features, including persistent digital identities, customizable avatars, and real-time communication channels, fostering social presence and collaboration at scale. To realize these experiences, reinforcement learning plays a vital role. RL enables agents to learn from interaction, making it ideal for dynamic, open-ended environments like the Metaverse where user behavior, goals, and world states are constantly evolving. Moreover, RL's ability to optimize sequential decision-making over time allows for smarter simulations and emergent behaviors, essential for sustaining immersive and responsive virtual worlds that evolve with user actions. As the Metaverse grows more complex and interactive, RL becomes a key enabler for creating scalable, intelligent systems that respond fluidly to human presence and intent. These capabilities make industrial Metaverse applications possible (Use-Case Box 1), where RL can optimize modules such as workforce training, process automation, and immersive simulations.

To explore the contribution of RL in the Metaverse applications and answer RQ1, this section explores the role of RL in supporting the various requirements of the Metaverse and enumerating the different classes of Metaverse problems solved through reinforcement learning. Figure 5 illustrates the distribution of the literature with regards to RL-based methods for the Metaverse. Naturally, the majority of the work proposes efficient solutions for the various communication challenges within Metaverse networks. The remainder of the literature proposes novel methods for virtual experiences and services which reside, virtually, in the Metaverse. Section 3.1 details the literature on RL solutions for Metaverse communication problems, exploring the role that RL agents play in empowering the enabler space of the Metaverse and analyzing the types of RL algorithms employed. Section 3.2 details the literature on the RL-based virtual experiences and services provided in the Metaverse.

3.1 The enabler space: efficient Metaverse communications

Creating an immersive experience surpasses simple browsing, and enabling users to live and interact within the virtual worlds requires sophisticated, cutting-edge infrastructure and intelligent algorithms to provide a higher Quality of Experience (QoE). With the advent of AR and VR technologies and edge computing, the user experience has been significantly improved (Vlahovic et al. 2022). However, utilizing such services in near-real time is challenging due to the "throughput requirement" and "the need for minimal latency." To address these challenges, many have proposed RL-based intelligent solutions to enable the immersive experience of Metaverse applications.

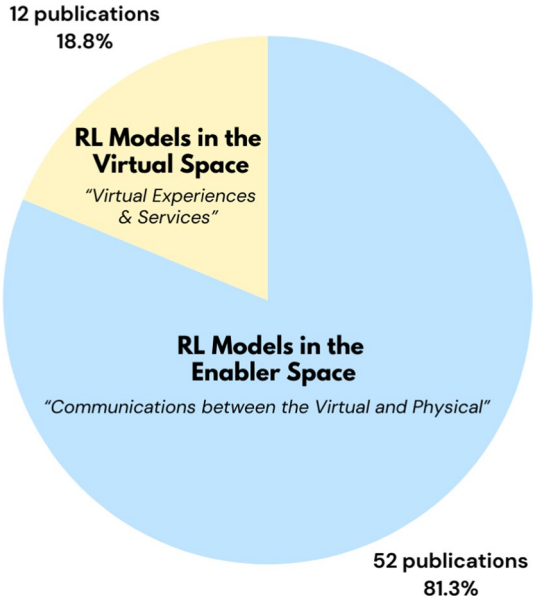

Fig. 5 The distribution of literature on the contribution of RL to support the Metaverse. The physical space is implicitly represented within the enabler space. The RL models optimizing the enabler space almost always optimize resources in the physical space with the goal of providing a more seamless virtual experience

3.1.1 Reinforcement learning algorithms in the enabler space

Reinforcement learning offers powerful tools for optimizing the allocation of computational, network, and storage resources for the large variety of Metaverse enabler networks. RL has evolved significantly in its application within Metaverse networks, reflecting advancements in algorithmic techniques and expanding problem domains. For instance, DQN, DDQN, PPO, A2C, and improved A3C have been proposed for multiple applications within the Metaverse. Figures 6 and 7 depict the categories and distribution of RL algorithms adopted for the resource management and synchronization in the enabler space of the Metaverse. Figure 6 focuses on the literature where one resource is optimized, whereas, Fig. 7 highlights those that jointly optimize multiple resource types. PPO, applied in an actor-critic architecture, has emerged as a dominant algorithm in Metaverse applications due to its robust performance and stability across diverse environments. Furthermore, DQN-based algorithms (Double DQN and Dueling DQN) and SAC are also widely adopted to enable the seamless immersion of Metaverse applications.

Along with the different RL algorithms, the architectures of the RL networks were studied. The majority of literature opts for a single value-based agent or single actor-critic network, where the RL agent has a single action space and is often making decisions over devices in the network in a centralized manner. Other works decentralize the action space, opting for multi-agent approaches, where each device or entity has an agent which is responsible for their respective decisions. Within the actor-critic approaches, some of the works employ *hybrid actors* i.e. multiple actors in a single network to be responsible for separate actions or *hybrid critics* i.e. separate critics, each with their own "perspective," evaluating a single agent. Interestingly, the RL algorithms applied to the enabler space in the Metaverse are the only ones within the scope of this review that employ hybrid-actor and hybrid-critic architectures.

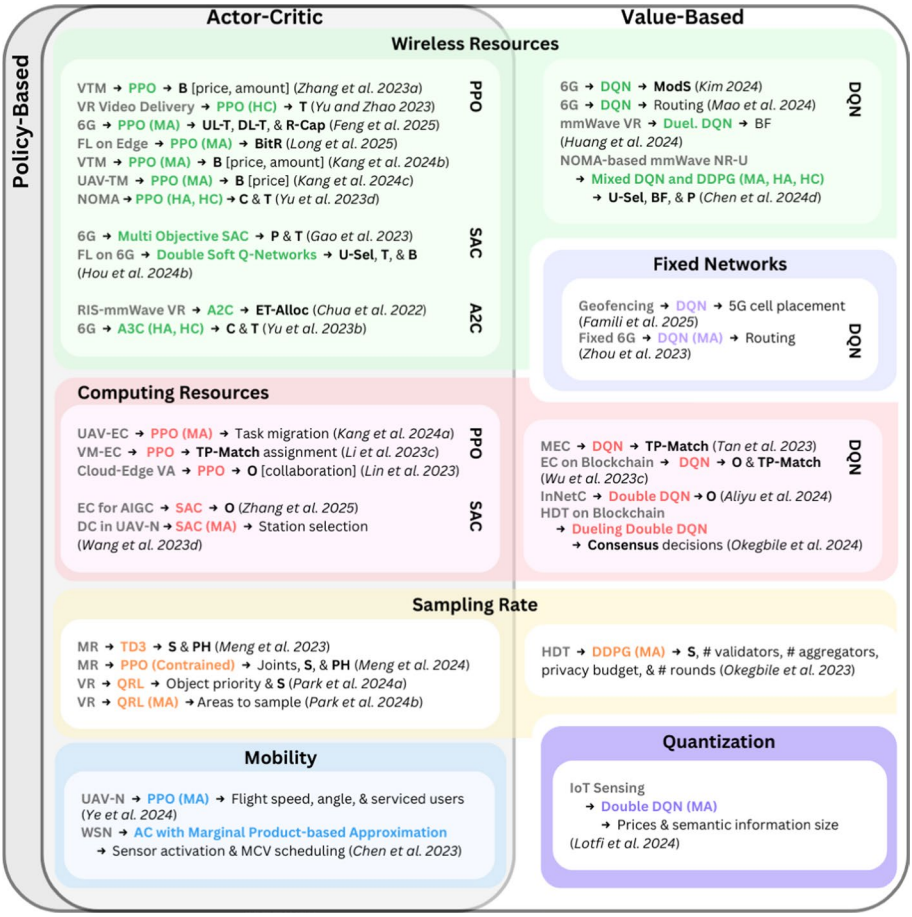

Fig. 6 The distribution of RL algorithms and architectures that support Metaverse enablers by performing resource management or synchronization tasks. **B**: bandwidth, **BF**: Beam Formation, **BitR**: Bit rate, **C**: channel allocation, **DC**: Data Collection, **DL-T**: downlink transmission, **EC**: Edge Computing, **ET-Alloc**: User Equipment-to-Tower allocation, **HDT**: Human Digital Twin, **InNetC**: In-Network Computing, **MCV**: Mobile Charging Vehicle,**ModS**: modulation strategies, **MR**: Metaverse Robotics, **NR-U**: New Radio in Unlicensed spectrum, **NOMA**: Non-Orthogonal Multiple Access, **O**: Offloading Strategy, **P**: phase angle, **PH**: prediction horizon, **R-Cap**: rendering capacity, **RIS-mmWave**: RIS-assisted mmWave for VR, **S**: Sampling rate, **T**: Transmission Power, **TP-Match**: Task-Processor matching, **UAV-EC**: UAV-Assisted Edge Computing, **UAV-N**: UAV-Assisted Networks, **VA**: Video Analytics, **VR**: Virtual Reality, **UAV-TM**: UAV Twin Migration, **UL-T**: uplink transmission, **U-Sel**: user selection, **VM**: Vehicular Metaverse, **VTM**: Vehicle Twin Migration, **WSN**: Wireless Sensor Networks

PPO-based approaches The majority of the literature on Metaverse communication problems employs PPO-based algorithms applied in an actor-critic architecture (Chen et al. 2024a; Chua et al. 2025, 2023; Kang et al. 2024b, c, a; Li et al. 2023c; Lin et al. 2023; Meng et al. 2024; Si et al. 2023a, b; Ye et al. 2024; Yu and Zhao 2023; Yu et al. 2023d, e, c; Zhang et al. 2023a). The popularity of PPO-based solutions in this domain is unsurprising due to its ease of implementation and its scalable and flexible nature. PPO is well suited for modern communication optimization problems since it performs well in high-dimensional and

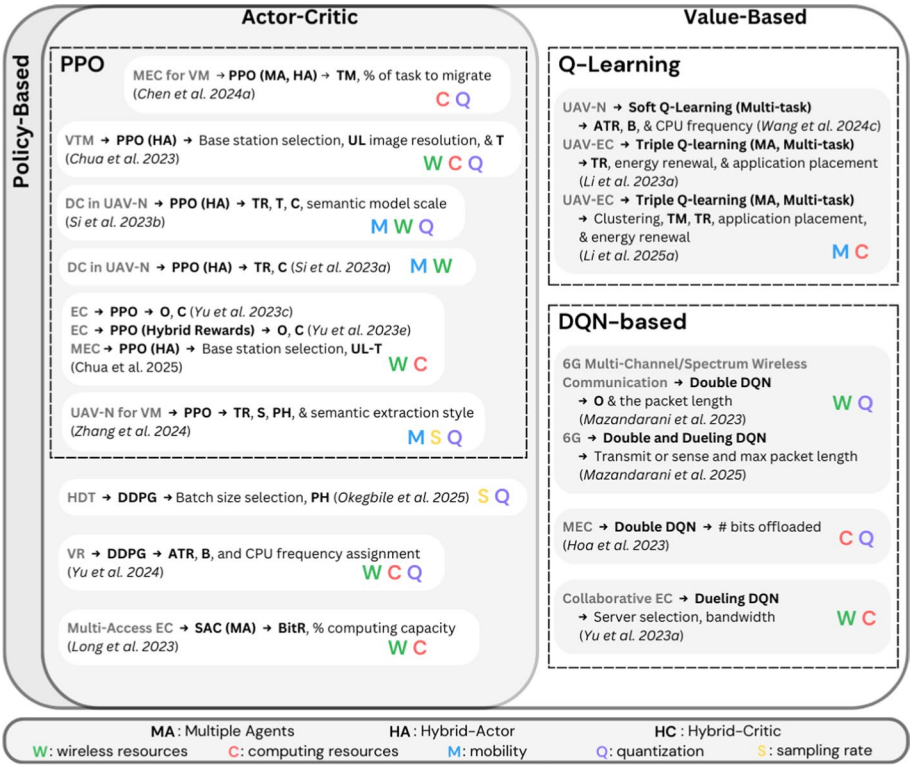

Fig. 7 The distribution of RL algorithms and architectures that support Metaverse enablers by performing joint resource management and synchronization tasks, highlighting the various architectures of RL agents employed. **ATR**: Attention-based tile resolution, **B**: bandwidth, **BitR**: Bit rate, **C**: channel allocation, **PH**: prediction horizon, **S**: Sampling rate, **T**: Transmission Power, **TM**: Task Migration, **TR**: Trajectory Planning, **UL**: uplink

continuous action spaces. In fact, PPO is adaptable to both discrete and continuous action spaces. PPO can be used for discrete decision making and resource management such as channel allocation (Si et al. 2023a, b; Yu et al. 2023d, e, c), device selection (Chua et al. 2023, 2025; Chen et al. 2024a; Li et al. 2023c; Lin et al. 2023; Si et al. 2023b), sampling rate selection (Meng et al. 2024; Zhang et al. 2024), bit rate change decisions (Long et al. 2025), and task migration or offloading decisions (Kang et al. 2024a; Yu et al. 2023e, c). It can also manage continuous resources such as transmission power (Chua et al. 2025, 2023; Si et al. 2023b; Yu and Zhao 2023; Yu et al. 2023d; Feng et al. 2025), bandwidth (Kang et al. 2024b, c; Zhang et al. 2023a), resolution selection (Chua et al. 2023), trajectory planning (Si et al. 2023a; Ye et al. 2024), and quantization levels (Chen et al. 2024a; Si et al. 2023b; Feng et al. 2025). Furthermore, PPO uses stochastic gradient descent (SGD) and adaptive clipping. This allows PPO to avoid complex second-order optimization and control the policy updates to avoid over-fitting; thus, making it computationally efficient and generalizable. They also allow PPO to be stable during training due to the introduction of a clipped objective function which makes PPO robust to excessively large updates to the policy function. These key

design features make PPO practical for a large variety of real-time, dynamic applications including Metaverse communication systems.

DQN-based approaches Another popular approach in the literature on Metaverse communications is DQN (Chen et al. 2024e; Kim 2024; Tan et al. 2023; Wu et al. 2023c; Mao et al. 2024; Famili et al. 2025) or DQN-based algorithms like Double DQN (Aliyu et al. 2024; Hoa et al. 2023; Lotfi et al. 2024; Mazandarani et al. 2023, 2025), Dueling DQN (Huang et al. 2024; Yu et al. 2023a; Mazandarani et al. 2025), or Dueling Double DQN (Okegbile et al. 2024). DQN comprises of deep neural networks to approximate the Q-function (Eq. 6), making it scalable to high-dimensional state spaces, which are common in Metaverse problems. Double DQN reduces overestimation bias by decoupling action selection and value evaluation, leading to improved stability in learning and better convergence. Dueling DQN breaks the Q-function into: the value function and the advantage function, where the former measures the importance of the current state and the latter learns the impact of the actions relative to the current state. Thus, Dueling DQN effectively identifies the most relevant actions to achieve faster convergence. Since DQN is designed for discrete action spaces, it has been applied in discrete decision making and resource management. Similar to PPO, DQN has been employed to perform channel allocation (Mazandarani et al. 2023), device selection (Tan et al. 2023; Wu et al. 2023c; Yu et al. 2023a; Mao et al. 2024), modulation type selection (Kim 2024), quantization level selection (Hoa et al. 2023; Mazandarani et al. 2023, 2025), antenna activation (Huang et al. 2024), geofencing 5G cell positioning (Famili et al. 2025), and task migration or offloading decisions (Aliyu et al. 2024; Wu et al. 2023c). In some cases, the decision variables were continuous, and thus, the action space was discretized in order to use DQN. For example, Lotfi et al. (2024) define a discrete set of action choices for the prices and semantic information sizes.

Multi-agent approaches While most of the literature explores RL solutions for Metaverse communication problems from a centralized perspective, some works choose to decentralize the decisions (Chen et al. 2024a, e; Kang et al. 2024b, c, a; Long et al. 2023; Lotfi et al. 2024; Wang et al. 2023e; Ye et al. 2024; Zhou et al. 2023; Feng et al. 2025; Long et al. 2025; Okegbile et al. 2023; Li et al. 2023a, 2025a). Decentralized decision-making systems are more scalable, robust, and fault tolerant. Adopting a multi-agent reinforcement learning approach in Metaverse communications can maintain operability even if some agents or nodes in the network fail. For example, Zhou et al. (2023) introduces a Cooperative Multi-Agent RL (Co-MARL) routing algorithm designed for the 6th Generation Fixed Networks (F6G), focusing on Metaverse applications with diverse Quality of Service (QoS) requirements. The Co-MARL approach allows multiple agents to collaboratively select optimal paths, effectively addressing the varying QoS demands by optimizing resource utilization across the network. Simulation results reveal that the Co-MARL achieves up to 23% higher rewards and consistently demonstrates better average latency, packet drop rate, and network throughput performance. In this routing example (Zhou et al. 2023), the failure of a router in the deployment stage can be easily rectified by rerouting the nodes adjacent to it, rather than re-computing the full path. Multi-agent approaches are ideal for distributed, large-scale systems such as Metaverse communication networks, allowing each agent to learn local decisions that contribute to a global objective without the bottlenecks or inefficiencies of centralized control. Long et al. (2023) integrated Graph Convolutional Networks (GCNs) with the Soft Actor-Critic (SAC) algorithm to optimize resource allocation in Multiaccess Edge Computing for the Metaverse. The proposed SAC-GCN models the Metaverse's mul-

tiuser environment as a graph, with each user represented by a node. GCNs capture user interactions, while the SAC algorithm dynamically allocates communication and computation resources. Experimental results in a virtual city park Metaverse scenario showed that SAC-GCN outperformed traditional methods, improving user experience by 27%, balancing resource allocation by 11%, and increasing resource utilization by 8%. Multi-agent reinforcement learning has been used to support cooperation (Chen et al. 2024a, e; Long et al. 2023; Wang et al. 2023e; Zhou et al. 2023; Ye et al. 2024; Feng et al. 2025; Long et al. 2025; Li et al. 2025a) and competition (Kang et al. 2024b) between the agents in resource management for enablers of the Metaverse. This is achieved through reward structures that incentivize the agents' cooperation, competition, or independent learning.

Hybrid-actor approaches In some cases, the solutions proposed in the literature choose to train multiple actors where each actor is responsible for a different decision and ultimately the joint decision is used (Chen et al. 2024a, e; Chua et al. 2025, 2023; Si et al. 2023a, b; Yu et al. 2023d, b). As opposed to the multi-agent approaches where each agent is performing the same type of action in a distributed manner, the hybrid-actor RL systems train actors to make unique actions and the joint decisions i.e. the combined individual actions of the hybrid-actors, are used. Hybrid-actor approaches allow for multiple decisions to be made especially when the action spaces of the decisions are different (discrete vs. continuous) (Chen et al. 2024a, e; Chua et al. 2025, 2023; Si et al. 2023a, b; Yu et al. 2023d, b). For example, the work of Chua et al. (2025) proposed a Multi-Agent Loss-Sharing model (MALS) utilizing DRL to optimize resource management for AR-based play-to-earn (P2E) games. The proposed approach minimized the downlink and uplink transmission latency, as well as worst-case battery expenditure while maximizing potential earnings influenced by user equipment resolution. The model integrates discrete-action RL for downlink User Equipment to Mobile edge computing Base Station (UE-MBS) allocation and continuous-action RL for uplink power selection. The proposed model achieved superior performance using multi-variable optimization analyses in complex optimization scenarios.

Hybrid-critic approaches In the literature, some RL-based solutions use multiple critics to train the RL agent(s) (Chen et al. 2024e; Yu and Zhao 2023; Yu et al. 2023d, b). Hybrid-critic approaches mitigate instabilities in the policy updates and help balance competing goals in multi-objective optimization. For example, Asynchronous Actors Hybrid Critic (AAHC), was proposed by Yu et al. (2023b) to solve the joint optimization of computation offloading, channel access arrangement, and downlink power allocation in a Metaverse 3D rending scenario. Unlike traditional MARL approaches, AAHC employs a hybrid critic to evaluate different stages and actions asynchronously, with distinct local and global objectives for two interacting RL agents. The method builds on the Advantage Actor-Critic structure, which separates action and value functions to improve stability and learning efficiency. Experimental results show that AAHC significantly outperforms baseline algorithms such as iterative RL (IteRL) and centralized RL (CRL) in terms of global reward, total delay, and retransmission counts.

Quantum reinforcement learning Recent advancements in quantum computing and simulation have shown great promise in enhancing synchronization and resource management in Metaverse systems through quantum reinforcement learning (QRL). Wang et al. (2023e) employ Quantum Collective RL (QCRL) to efficiently manage UAVs that capture and transmit images to base stations for rendering in the Metaverse. It allows UAVs to adapt swiftly to changing environments, ensuring seamless synchronization and efficient task

offloading to Web3 cloud servers when resources are limited. The authors show that their proposed approach enhances offloading efficiency and overall system performance. Park et al. (2024a) proposed a novel spatio-temporal prioritization system for optimizing synchronization in the Metaverse. The authors use quantum RL (QRL) algorithms to address the NP-hard problems of spatial and temporal prioritization. The QRL approach leverages quantum neural networks (QNNs) for high scalability and reduces control dimensions to a logarithmic scale. Extensive experiments with realistic datasets validate the algorithm's effectiveness, demonstrating its superiority over classical RL methods in enhancing user experience and system performance. Park et al. (2024b) proposes another spatio-temporal syncronization system using QRL. The authors propose a novel coordination algorithm, Quantum Multi-Agent Reinforcement Learning (QMARL), to achieve high synchronization between physical space and virtual metaverse space, by scheduling the sampling of data from the physical space based on the Age of Information (AoI). Collectively, these studies highlight the potential of QRL-based methods in addressing key challenges of synchronization, scalability, and real-time responsiveness in complex Metaverse environments.

3.1.2 The role of RL in the enabler space

As illustrated in Fig. 2, the enabling networks and technologies of the Metaverse facilitate seamless interaction between the virtual and physical worlds through two main activities: *resource management* and *synchronization*. Resource management encompasses the efficient allocation of communication channels, storage, computational power resources, and physical positions of the enablers. Synchronization activities go hand-in-hand with the existing resource management activities while adding concerns regarding the efficient collection of data, the size of information to transfer, and how often to transfer it. In the literature, RL approaches have been proposed to perform the different resource management and synchronization tasks in the Metaverse enabler space with the goals of minimizing latency, maximizing throughput, and maximizing the quality of experience of the user. These tasks include: a) *wireless resource management*, b) *computing resource management*, c) *mobility management*, d) *quantization*, and e) *sampling*. The majority of the literature proposes to solve the latency and throughput problems though joint optimization of the aforementioned tasks. The literature focuses greatly on the management of wireless and computing resources with a few papers proposing RL-based solutions exclusively for optimizing sampling rates, the mobility of agents in the network, or the quantization of the data transferred.

Wireless resource management The literature explores RL-based solutions for optimizing communications through the efficient allocation of wireless resources such as channel access (Yu et al. 2023d, b; Si et al. 2023a; Mazandarani et al. 2023, 2025; Si et al. 2023b; Yu et al. 2023e, c), transmission power (Yu et al. 2023d, b; Yu and Zhao 2023; Si et al. 2023a, b; Chua et al. 2023, 2025; Feng et al. 2025), bit rate (Long et al. 2023, 2025), and bandwidth (Zhang et al. 2023a; Hou et al. 2024b; Kang et al. 2024b, c; Yu et al. 2024, 2023a; Mao et al. 2024) decisions for the transfer of data between devices in the network of Metaverse enablers. The work of Chua et al. (2022) uses deep reinforcement learning (DRL) to decide the optimal matching of user equipment to Metaverse cell stations given the current channel gain between the users and cell towers and the transmission power of the cell stations. Kim (2024) uses DRL to control the modulation order for the 3GPP dynamic transmission strategy in the emerging 6 G networks. Some works in the literature propose to optimize Meta-

verse communication using RL-based bandwidth allocation or pricing (Zhang et al. 2023a; Hou et al. 2024b; Kang et al. 2024b, c; Yu et al. 2024, 2023a). More specifically, Zhang et al. (2023a), Kang et al. (2024b), and Kang et al. (2024c) model the resource allocation for digital twin migration using Stackelberg games between Metaverse service providers and the user vehicles or UAVs. RL-based solutions were also proposed for Metaverse VR wireless resource management for non-orthogonal multiple access (NOMA) based millimeter wave (mmWave) enabled with reconfigurable intelligent surfaces (RISs). For example, Gao et al. (2023) uses an SGD algorithm to localize the users and consequently uses RL to perform updates to the phase angle and transmission power. Huang et al. (2024) focused on the adaptive configuration of the mmWave antennas of the VR headsets to reduce the drops in transmission rate caused by the user's movement. The proposed Dueling DQN-based algorithm configures the receiver's phase shift and adjust the beam width as the user's viewpoints vary. The proposed framework uses viewpoint prediction to estimate users' head movements and learns to avoid beam misalignment and control the transmission capacity, thus minimizing power consumption while meeting transmission requirements. Chen et al. (2024e) dynamically controls the Metaverse enabler network using a multi-agent hierarchical deep reinforcement learning (MAHDRL) algorithm that performs user selection and coordinates beamforming among base stations to overcome bandwidth limitations.

Computing resource management Other works in the literature propose RL-based solutions to reduce latency through efficient computation offloading, often leveraging an existing network of edge devices. The RL agents in these solutions are tasked with learning to make efficient decisions on whether or not to offload computational tasks (Wang et al. 2024c; Chen et al. 2024a; Hoa et al. 2023; Yu et al. 2023e), which devices to offload the tasks to (Chua et al. 2023; Yu et al. 2023a; Chua et al. 2025; Yu et al. 2023c; Zhang et al. 2025; Li et al. 2023a; Okegbile et al. 2024; Li et al. 2025a), and the computational power of the devices assigned to the tasks (Yu et al. 2024; Long et al. 2023). RL-based solutions were proposed with a focus on these discrete decisions in the literature (Wang et al. 2023e; Aliyu et al. 2024; Li et al. 2023c; Lin et al. 2023; Wu et al. 2023c; Tan et al. 2023; Kang et al. 2024a). Some of the literature additionally optimizes the offloading strategy (Aliyu et al. 2024; Lin et al. 2023; Wu et al. 2023c). A work done by Aliyu et al. (2024) shows that the RL-based DDQN method effectively reduces overall system costs and enhances resource utilization by dynamically optimizing offloading decisions. Here, the RL agent determines the best out of three strategies, which are: local computing, offloading to fog in-network computing nodes (FINs), or offloading to edge in-network computing nodes (EINs). Compared to traditional methods like Mobile Edge Computing (MEC), the proposed approach achieves up to a 70% reduction in costs and a 16% increase in system rewards represented by the the difference between the total current cost and total cost resulting from the partial offloading policy of the system. Wu et al. (2023c) proposed blockchain-enhanced intelligent computing framework, BC-MetaCast, to address the challenges of Metaverse livecasting, particularly its compute-intensive and delay-sensitive nature. The proposed framework uses edge servers as cluster head and leverages user equipment to perform offloading in a decentralized manner. It integrates an On&Off-Chain Superimposed DRL (Sup-DRL) algorithm for effective computational task scheduling. The authors show that BC-MetaCast outperforms state-of-the-art schemes in terms of high concurrency, efficient processing, and service quality, where it showed superior convergence and performance in handling dynamic environments and large-scale user interactions.

Mobility management In some Metaverse networks, the mobility of the users devices as well as the mobility of the network edge devices are considered. Some of the literature considers UAV-assisted Edge Metaverse networks where the UAVs are deployed to collect and transfer real-world data (Si et al. 2023a, b; Ye et al. 2024; Zhang et al. 2024; Chen et al. 2023), or perform offloading (Wang et al. 2024c; Chen et al. 2023; Li et al. 2025a). In these scenarios, RL-based algorithms were developed to optimize the trajectories of the UAVs in performing their respective tasks, while minimizing mission time (Si et al. 2023a, b; Ye et al. 2024; Wang et al. 2024c; Zhang et al. 2024; Chen et al. 2023; Li et al. 2025a). Si et al. (2023a) propose PPO-based RL agents to optimize channel allocations and UAV trajectories in UAV-assisted Metaverse network scenarios. Utilizing RL shortened mission completion times while efficiently collecting data from roadside units (RSUs) across urban environments. Si et al. (2023b) proposed a hybrid RL framework to enhance uplink semantic communications for Metaverse users in remote areas, with UAV assistance. It employs two RL agents: one for discrete actions (channel allocation) and another for continuous actions (transmission power, semantic model scale, and UAV trajectory). This dual-agent approach leverages PPO to optimize the combined action. Simulation results demonstrate the effectiveness of this hybrid action RL framework. Compared to benchmarks like Equal Power Allocation (EP) and a triple PPO agent approach, the proposed method significantly reduces data collection time and improves efficiency, especially as the number of users increases. Ye et al. (2024) proposed a framework using MADRL to maintaining high-quality information (QoI) by optimizing the trajectories of UAVs acting as mobile base stations. This approach schedules the motion dynamics of the UAVs to ensure efficient data collection from mobile users equipped with Metaverse devices, even in dynamic and changing environments. Additionally, Wang et al. (2024c) proposed the MURAL (multitask DRL-based two-stage alternating) model to jointly allocate UAV charging time, schedule computation tasks, and optimize UAV and device trajectories to maximize system computation efficiency under energy constraints. Utilizing a Manhattan city map simulation, theoretical analysis, and performance evaluations demonstrate that MURAL significantly outperforms existing methods in convergence speed and computation efficiency.

Quantization In addition to the optimization of Metaverse communications through the intelligent allocation of available resources, the literature also proposes to optimize the size of the the information transferred, especially in scenarios when the network resources are limited. The use of RL-based approaches has been proposed to determine: (1) the portion of tasks to offload in vehicular Metaverse edge computing networks (Chen et al. 2024a), (2) the quantity of bits to offload to the edge server (Hoa et al. 2023), (3) the size and quality levels of information to gather from the physical world and present in the virtual space (Lotfi et al. 2024; Si et al. 2023b; Chua et al. 2023; Yu et al. 2024; Zhang et al. 2024; Okegbile et al. 2025), and (4) the packet length in wireless communications through multi-channel spectrum wireless communications (Mazandarani et al. 2023, 2025). Hoa et al. (2023) address the latency requirements of data gathered by UAVs utilized by Metaverse Virtual Service Providers (VSP). The VSP can locally process some data while offloading the rest to an edge server (ES). However, the challenge lies in optimizing the data offloading to ensure timely Metaverse services while managing the uncertainties of data collection, channel conditions, and computing resources. To address this, the authors model the offloading as a stochastic problem and apply DDQN model to optimize the number of bits to offload. Lotfi et al. (2024) address the challenges of limited bandwidth faced by VSPs when hiring sensing IoT

devices to offer their sensing data for the creation and rendering of virtual digital twins of the physical world. Although the use of semantic extraction algorithms can reduce the size of the data to obey bandwidth constraints, the consequent asymmetry of information between the VSPs and the IoT devices results in a multi-dimensional adverse selection problem. Lotfi et al. (2024) propose the use of iterative contract design and multi-agent reinforcement learning (MARL) with Prioritized Double Deep Q-Learning (PDDQL) in a semantic aware truthful digital twin synchronization mechanism for the Metaverse. The authors design a framework with 2 layers which operate iteratively: the uplink or *upstream* layer between the IoT sensing devices and the VSP, and the downlink or *downstream* layer between the VSP and the end-users in the Metaverse. In the upstream layer, the RL agent determines the prices and the semantic information size for each contract. In the downstream layer, the RL agent decides the quality of the semantic information rather than the size. Chua et al. (2023) address the issue of adversarial attacks on AR-assisted driving systems, where adversarial patches placed on physical objects can distort the Metaverse Map. The authors propose a deep reinforcement learning-based Heterogeneous Action (HA) algorithm to optimize the allocation of AR vehicles to Metaverse Map Base Stations (MMBS), the resolution of physical world scenes for upload, and the uplink transmission power. Extensive experiments demonstrated the HA model's superiority in rewards, transmission latency, battery consumption, and mean Average Precision scores, especially in complex scenarios with multiple AR vehicles. Yu et al. (2024) present an online RL algorithm for resource management in MEC-enabled VR content streaming which relies on the dynamic quality of experience (QoE) metric. They define QoE using a combination of latency and attention-based resolution in order to efficiently allocate CPU and bandwidth resources.

Sampling The challenge of synchronizing the real world and the virtual world requires seamless connection as well as efficient decisions on how often the state of real entities should be collected and reflected by their digital twins. In the literature, RL algorithms were used to select the sampling rate and its prediction horizon to solve synchronization problems in Metaverse applications (Meng et al. 2023, 2024; Park et al. 2024a; Zhang et al. 2024; Okegbile et al. 2025). Meng et al. (2023) address the challenge of synchronizing real-world devices with their digital counterparts in the Metaverse by proposing a novel co-design framework for sampling, communication, and prediction. The framework is optimized using a Knowledge-assisted Constrained Twin-Delayed Deep Deterministic (KC-TD3) algorithm, which integrates expert knowledge with advanced DRL techniques. This approach aims to reduce communication load while adhering to a Mean Squared Error (MSE) constraint between the real and digital models. The experimental validation, conducted with a real robotic arm and its virtual model, demonstrates that the proposed framework significantly outperforms traditional methods. It reduces communication load by up to 87% and improves tracking error. The KC-TD3 algorithm improves convergence time and stability and adapts dynamically to communication conditions such as varying packet loss probabilities and prediction horizons. Later, Meng et al. (2024) presents a task-oriented cross-system design framework to optimize the modeling of a real-world robotic arm in the Metaverse. The framework integrates sensing, communication, prediction, control, and rendering to minimize the packet rate required for timely and accurate modeling. A Constraint Proximal Policy Optimization (C-PPO) algorithm, derived from PPO, is introduced to optimize scheduling policy and prediction horizons. The C-PPO algorithm incorporates domain knowledge, such as the Jacobian matrix for motion analysis and the Conditional Value-at-

Risk (CVaR) for modeling error constraints. A prototype system, featuring a real-world robotic arm and its digital counterpart in the Metaverse, was built to evaluate the proposed methods. Experimental results demonstrated that domain knowledge integration reduced convergence time and packet rate by up to 50%, outperforming a baseline framework in packet rate efficiency and modeling error distribution.

3.2 The virtual space: immersive experiences and reliable services

The integration of RL within the Metaverse enhances user experiences and autonomy in virtual environments. By leveraging RL, the Metaverse can offer personalized user experiences, adaptive content generation, and intelligent agent behavior. In fact, reinforcement learning is emerging as a common "decision engine" across otherwise dissimilar virtual Metaverse tasks. To explore the diversity in the benefits of RL for the virtual space of the Metaverse, this section compares the literature (summarized in Table 3) through the lens of how RL is used to further the goals of Metaverse systems such as trust, immersion, and realism. Subsequently, the measures of success for each goal are analyzed; namely, the quantification of user experience and satisfaction. With a limited number of articles leveraging RL for virtual Metaverse applications, the virtual integration of RL remains relatively underexplored, leading to a discussion surrounding the transferability of the existing work and an analysis of gaps such as, human or human-aware feedback mechanisms, the interpretability of agent decisions, and the scalability of RL solutions across heterogeneous platforms.

3.2.1 Cultivating trust

Trust in the Metaverse virtual space would be most affected by a lack of security or service disruption. Metaverse users should trust that 1) their data is secure and 2) their requests will be reliably and effectively fulfilled. In recent years, RL has been proposed as solution for the security and management of virtual services in the Metaverse. From a security perspective, an interesting application was proposed by Li et al. (2023b) where an RL-based optimization method was designed to enhance the mutation scheduling process in fuzzing to discover software vulnerabilities in the Metaverse. Traditional fuzzing methods often suffer from randomness and inefficiency in generating test cases, which reduce their effectiveness in uncovering software vulnerabilities. The proposed AFL-RL model addresses this issue by applying an improved TD3 RL algorithm in American Fuzzy Lop (AFL), a fuzzing software developed by Michal Zalewski.[4] Experimental results show the superiority of the proposed AFL-RL algorithm compared to traditional AFL techniques by achieving better path coverage and mutation efficiency across multiple software, including image and video processing tools like libtiff and ffmpeg. With the success of AFL-RL in fuzzing, other RL–based approaches can be applied to improve Metaverse security and service reliability. For instance, RL can be leveraged for intrusion detection systems (IDSs) that dynamically adapt to evolving cyberattacks. By continuously learning from network traffic and user behavior, RL-driven IDSs can identify anomalies such as denial-of-service (DoS) attacks or data exfiltration attempts more efficiently than static rule-based systems.

Another aspect of trust is ensuring that user requests for services such as VR video, audio, or AI generated content are reliably accepted (Chu et al. 2024; Du et al. 2024),

[4] American Fuzzy Lop: http://lcamtuf.coredump.cx/afl/.

Table 3 Metaverse Virtual Experience and Services

References	Domain	Task	RL algorithm	Multi-agent	Simulator
Xu et al. (2022)	VR/AR/XR	Requester-Provider Matching	Double Dutch Auction PPO Auctioneer	✗	–
Chu et al. (2024)	VR/AR/XR	Requester-Provider Matching	Dueling Double DQN on SAC	✗	–
Du et al. (2024)	AI-Generated Content	Requester-Provider Matching	SAC (D2SAC)	✗	–
Jin et al. (2022)	Virtual Concert	Music Generation	DQN (TransformerXL + DQN)	✗	Unreal Engine
Lin et al. (2022a)	Metaverse Games	NPC Design	PPO (PPO + ICDVAE)	✗	Open AI Gym
Lin et al. (2022b)	Metaverse Games	NPC Design	SAC (Fed-MT-ISAC)	✗	MujoCo and Meta-World
Hare and Tang (2023)	Education	NPC Design	Double DQN (Hierarchical)	✓	Gridlock
Pu et al. (2024)	Automated World Design	Rule Generation	RGN with RL as an evaluator (Q-learning)	✓	Python and Unity
Zhang et al. (2023b)	Video Streaming	Immersion (Video)	A3C	✗	–
Chen et al. (2024c)	Human Digital Twin	Immersion (Video)	Double DQN (Conditional Diffusion Actor)	✗	–
Wu et al. (2025)	Human Digital Twin	Immersion (Video & Haptic)	PPO	✗	Unity, Chai3D
Li et al. (2023b)	Software Security	Fuzzing	TD3	✗	libtiff & ffmpeg

completed with high quality (Du et al. 2024). Chu et al. (2024) proposed MetaSlicing to address the significant resource demands of the Metaverse, such as computing, storage, and networking resources to support high-speed and low-latency connections. The proposed MetaSlicing framework manages and allocates resources by grouping applications into clusters called MetaInstances, where shared functions improve resource utilization. The authors utilized RL to handle dynamic and uncertain resource demands. Specifically, an intelligent admission control algorithm based on a semi-Markov Decision Process (sMDP) and Deep Reinforcement Learning (DRL) is developed to make real-time decisions about resource allocation, optimizing the system's performance and maximizing long-term revenue. Chu et al. (2024) proposed an RL-assisted novel resource management framework to allocate resources across different tiers for Metaverse applications dynamically. The proposed MetaSlicing approach groups the applications into MetaInstances to share common functions, and iMSAC (intelligent MetaSlice Admission Control) is used to learn the optimal admission policies. The authors show that MetaSlicing significantly outperforms greedy policies, enhancing long-term revenue by up to 80% and request acceptance probability by 47%. DRL enables the system to learn and adapt to changing resource demands and user behaviors, ensuring efficient and effective resource management. Another example of RL for matching Metaverse virtual service providers with requesters comes from Du et al. (2024) who proposed an AI-generated content-as-a-service (AaaS) architecture based on AI-generated Content (AIGC) models in wireless edge networks to enhance accessibility and provide personalized user experiences. The authors utilized diffusion model-based AI-Generated Optimal Decision (AGOD) algorithm to optimize the selection of AIGC Service Providers (ASPs) under uncertainty. To further enhance ASP selection, the AGOD is integrated with DRL as the Deep Diffusion Soft Actor-Critic (D2SAC) algorithm. The proposed D2SAC demonstrates superior performance compared to seven existing DRL algorithms (iDQN, DRQN, Prioritized-DQN, Rainbow, REINFORCE, PPO, and SAC) in terms of cumulative reward, training time, and convergence speed. D2SAC achieved the highest training and test rewards, with a fast convergence rate and efficient training time. Du et al. (2024) defined the rewards using the quality of the generated content as well as a penalty for crashes (when a service is not assigned). The quality of the generated content is determined using a human-aware content quality function, Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), trained to predict image quality based on the perception of humans.

Both Chu et al. (2024) and Du et al. (2024) focused on the user-experience as a metric; conversely, Xu et al. (2022) chose to optimize social welfare in VR service delivery. Xu et al. (2022) introduce an innovative economic framework designed to optimize the delivery of VR services within a wireless edge-empowered Metaverse. Unlike traditional continuous trading markets, they propose a Double Dutch Auction (DDA) mechanism tailored for dynamic matching and pricing of VR services between users and service providers (SPs). To improve the efficiency of their DDA, the authors integrate DRL, where the auctioneer acts as a DRL agent that learns optimal auction policies without prior knowledge. Their proposed approach aims to maximize social welfare by minimizing regret for the buyers and sellers as well as minimizing communication costs associated with auction information exchange. The system focuses on non-panoramic VR applications delivered via wireless networks, emphasizing immersive experiences such as interactive games and virtual concerts. Simulation results validate the effectiveness of their approach across different market sizes and VR service requirements. The DRL-based DDA shows robust convergence

under varying market scales, although larger markets require more epochs for stabilization. Comparative analysis against baseline methods demonstrates superior performance in terms of social welfare optimization and reduced auction information exchange costs, highlighting its potential for efficient VR service delivery in wireless edge-empowered Metaverse environments.

3.2.2 Maintaining immersion

As evidenced by the plethora of research within the enabler space (Sect. 3.1), immersion is viewed as the highest goal of the virtual space in Metaverse systems. While in Sect. 3.1 the research focus is the management of resources and seamless synchronization between the physical and virtual worlds using RL, this section takes a look at the role of RL in managing the visual aspects of immersion. More specifically, it highlights how RL can be used to select the quality of the media presented in the virtual space in order to optimize the trade-off between quality and resource load. In the literature, RL agents learn to select the bitrate of video content both overall (Chen et al. 2024c) and tile-wise (Zhang et al. 2023b; Wu et al. 2025). Chen et al. (2024c) explores how mobile AI-generated content (AIGC) can enable Human Digital Twins (HDTs) in personalized healthcare and proposes a Conditional Diffusion Model-based approach (CODI) to optimize resource allocation and QoE for Human Digital Twin (HDT) applications. CODI works by first generating random decisions (e.g., each user's resolution ratio and diffusion step) from a Gaussian distribution, then iteratively refining them through a multi-step denoising process to yield optimal decisions that maximize users' total QoE. It is trained under an actor-critic deep reinforcement learning framework, where the diffusion model acts as the actor to produce decisions, and the critic (using double Q-learning) evaluates their value. Compared with conventional DRL algorithms SAC and PPO, CODI demonstrates faster convergence, better stability, and higher final rewards in training. It also achieves superior performance when scaling with the number of users, due to its stronger sampling quality and long-term dependency handling. Chen et al. (2024c) models QoE as a linear combination of the VR video bitrate and the similarity between actual video stream and generated one. Similarly, Wu et al. (2025) focuses on Human Digital Twins (HDTs) and the challenge of transmitting multi-modal signals (visual and haptic) between physical twins (PTs) and virtual twins (VTs) to ensure immersive experiences. The authors design a new Quality of Experience (QoE) model that jointly considers video quality, switching rates, and haptic feedback errors. They formulate the transmission as an optimization problem constrained by limited bandwidth and the need for synchronization between modalities. Wu et al. (2025) propose VisHap, a deep reinforcement learning–based online optimization algorithm that dynamically allocates bandwidth, bitrate, rendering modes, and compression thresholds. Experiments using an HDT multi-modal interaction platform show VisHap outperforms benchmarks, ensuring synchronized, high-quality immersive interactions. Similarly, Zhang et al. (2023b) provides a unique methodology for optimizing video bitrate using RL to maintain high QoE under network constraints. Zhang et al. (2023b) proposed AI and Blockchain within a Metaverse framework (Web 3.0, AIB-Metaverse) which addresses the limitations of traditional World Wide Web (WWW) inefficiencies such as privacy concerns, and low Quality of Experience (QoE). AI in Web 3.0 enables personalized experiences and efficient decision-making, while Blockchain ensures decentralization and data ownership protection. The study focuses

on AI's role in enhancing Metaverse video delivery, specifically through a Field of View (FoV)-based approach to reduce bandwidth waste and optimize user QoE. The proposed AI-based approach employs a Dynamic RL model for adaptive bitrate selection in Metaverse video streaming. It predicts user FoV to determine necessary video tiles and their bitrates, optimizing video quality within bandwidth constraints. The system utilizes a Hidden Markov Model for FoV prediction and a 1D-CNN for state processing, training on open-access network trace datasets (4 G-LTE, HSDPA, broadband). Comparative experiments against heuristic and existing DRL methods demonstrate superior QoE performance of the proposed AI model; where QoE is defined using the average bitrate of tiles, and penalties for variation between tiles in the FoV, variation between neighboring tiles, and rebuffering time. The authors show that AIB-Metaverse-based Web 3.0 represents a significant step toward a more efficient, secure, and immersive internet experience.

3.2.3 Content generation and realism

Content generation is the most creative and diverse field of virtual Metverse research, spanning the production of visual assets, immersive virtual experiences, and game environments. A central objective in this field is achieving realism: generated content must maintain coherence with social norms and avoid disruptions that could "break" immersion. Reinforcement learning has been applied in several branches of this domain, Non-Playable Character (NPC) design (Hare and Tang 2023; Lin et al. 2022a, b), music generation (Jin et al. 2022), and world design (Pu et al. 2024).

Lin et al. (2022b) addresses the construction of human-like NPCs using imitation learning in Metaverse games, where the complexity and diversity of experience leads to sparse data due to low interoperability. Some approaches to solving the problem lead to privacy issues and knowledge transfer complications. Thus, the authors propose to extend the Inverse Soft Actor-Critic (ISAC) to federated learning and multi-task problems, termed Federated Multi-Task Inverse Soft Actor-Critic (Fed-MT-ISAC). The performance of the proposed algorithm is compared to Policy Distillation Transfer (PDT) and Dynamic-Aware Variational Transfer (DAVT) in multi-style tasks and multi-skill tasks, where empirical results show that Fed-MT-ISAC improves performance in all experimental settings such as the Mujoco task "Ant" and robotic manipulations in Meta-World. Lin et al. (2022a) address the challenges of designing the reward functions for large-scale RL-based non-player characters (NPCs) in the complex and highly dynamiuc environments of the Metaverse. The authors propose a curiosity mechanism, named Intrinsic Curiosity-Driven Variational Auto-Encoder (ICDVAE) for RL-based NPCs which is distribution-aware. It learns to construct the state distribution in order to represent novelty by integrating a curiosity mechanism with a generative model. The method outperformed baselines in complex Atari video games, including "Breakout", "SpaceInvaders", "KungFuMaster" and "Qbert", and under multiple environments. Since both Lin et al. (2022b) and Lin et al. (2022a) aim at building game NPCs, they evaluate the RL-based NPCs by tallying the rewards gained in-game or at task completion. For more specific and complex applications, such as online training or education, Hare and Tang (2023) proposed an improved methodology for Metaverse learning through a hierarchical multi-agent RL (HRL) approach with experience sharing, specifically targeting AI-based NPCs to personalize educational experiences. The framework utilizes binary decision trees to model learner behavior and guide NPC interactions based on real-world learner

data. Each agent's actions are incentivized based on learner performance variables, such as test scores, time management, and emotional responses captured via webcam. Results from extensive simulations demonstrate that the hierarchical approach outperforms single-agent strategies in terms of cumulative rewards and episode completion steps. Moreover, experiments comparing different experience sharing methods highlight the efficacy of weighted experience sharing, showing significant improvements in agent learning and overall system performance.

Content generation for the Metaverse is an open area for research due to the dynamic and vastly connected nature of Metaverse systems. Jin et al. (2022) emphasized enhancing the musical and auditory aspects alongside visual immersion in the Metaverse. They proposed a comprehensive framework for Metaverse concert generation using RL and neural networks to improve music generation and spatial audio rendering. The authors improved the Transformer-XL music generation network Dai (2019) using Monte Carlo and DQN methods to optimize music generation and tonal quality. Experimental results on the POP909 dataset Wang et al. (2020) demonstrate significant advancements over traditional methods like Melody_LSTM and RL-tuner regarding objective metrics such as pitch stability, polyphony, and tonality. Additionally, the study introduces a novel neural rendering approach for spatial audio synthesis, leveraging a fully convolutional neural network (CNN) for accurate binaural audio generation. It outperformed conventional techniques like Head-Related Transfer Function (HRTF) regarding reliability and fidelity for creating immersive auditory experiences in virtual concert settings. Furthermore, the authors propose a metadata-based audio rendering algorithm for simulating realistic acoustic environments within the Unreal Engine 4 (UE4). This system complements the spatial audio capabilities by reconstructing sound fields tailored to virtual concert stages, enhancing the overall auditory immersion. Subjective evaluations involving professional and non-professional listeners affirm the superiority of the proposed Transformer-XL variants and neural audio synthesis methods in delivering enriched musical experiences within the Metaverse. Hence, integrating RL models with generative neural networks is beneficial in enhancing the user experience in the Metaverse.

World and game design in the Metaverse presents a need for evolutionary studies of the effect of governance decisions. Pu et al. (2024) proposed Rule Generation Networks (RGN) framework for automated rule design, evaluation, and evolution based on generative neural networks and RL models to create and refine rules within simulated environments. The RGN aims to replicate the dynamics of natural life through computational models, focusing on the adaptive nature of rules in various domains. By formalizing the elements of rules, strategies, and evaluation, the framework seeks to elucidate the intricate relationships involved in rule-generation tasks. The paper used two digital environments, Maze Run and Trust Evolution,[5] to demonstrate the framework's capabilities. These environments are designed using Python and Unity and serve as platforms for both AI (RL NPC models) and human participants to interact and evaluate the generated rules. The RGN framework incorporates a controllability metric to gauge game dynamics and evolve the rule designer accordingly. In the Maze Run environment, agents successfully navigated increasingly complex mazes as the rules evolved, indicating the framework's capability to produce adaptable and effective rules. In the Trust Evolution environment, the framework modeled social dynamics and trust, providing insights into how rules influence social behaviors and interactions. The results

[5] Trust Evolution: https://ncase.me/trust/.

demonstrated that the RGN framework effectively generates and refines rules, adapting to changes and improving performance over time.

3.2.4 Transferability and the human perspective

In the surveyed literature, a recurring theme is that RL solutions are often designed with transferability in mind, requiring minimal adaptation to suit the diverse and evolving digital ecosystems of the Metaverse. General RL frameworks can thus be reused across multiple contexts, while more specialized approaches target narrower domains. For instance, hierarchical multi-agent RL has been applied in education to coordinate intelligent non-player characters (NPCs) and learning companions (Hare and Tang 2023). Such techniques are not only transferable within the educational space but also hold promise for other complex human-centric tasks, such as dynamic traffic moderation in virtual cities (Li et al. 2025b) or adaptive healthcare support within human digital twins (Chen et al. 2024b).

Despite this potential, most studies rely on predefined or simulated quality-of-experience (QoE) metrics to evaluate their systems. Only a limited number integrate human evaluators to directly assess realism, engagement, or usability (Pu et al. 2024; Jin et al. 2022; Wu et al. 2025). The use of RL as a surrogate for human behavior in Metaverse game design illustrates both the practicality and creativity of this approach: it allows researchers to approximate costly and time-consuming user studies while still generating rules and dynamics that reflect plausible human decision-making (Pu et al. 2024). Similarly, specialized use-cases such as RL-driven fuzzing (Li et al. 2023b) demonstrate how RL can augment or replace metaheuristics (e.g., genetic algorithms) by adaptively modeling their mutation, crossover, or branching strategies to reduce inefficiency.

Looking ahead, future research should explore hybrid frameworks where human feedback and RL agents co-evolve, enabling systems that learn from both simulated environments and lived user interactions. Cross-domain generalization of RL policies, where solutions developed for education, gaming, or healthcare seamlessly extend into adjacent domains, remains a key challenge. Equally important is the development of richer evaluation metrics that go beyond technical performance (e.g., latency or accuracy) to capture subjective aspects such as user trust, perceived fairness, and long-term engagement. By embedding the human perspective more deeply into the design loop, RL-driven Metaverse systems can move closer to being adaptive, trustworthy, and genuinely user-centric.

4 Metaverse RL applications for real-life

The Metaverse offers a comprehensive virtual environment for training RL algorithms where RL agents can undergo extensive training in a controlled, simulated setting that closely mimics real-world conditions. This approach enables the development of robust and adaptable policies that can then be transferred to real-world applications with minimal fine-tuning. This section addresses RQ2 by exploring the utilization of Metaverse applications for real-life decisions. The dynamic virtual environments allow RL models to learn complex behaviors and strategies that translate effectively to real-world scenarios, thereby reducing the time and cost of real-world training and experimentation. Section 4.1 discusses the works done to bridge the gap between simulation and reality, showcasing how these

virtual environments can inform practical applications followed by Sect. 4.2 that explores the role of human feedback in shaping RL learning processes, emphasizing the relationship between human intuition and machine-driven algorithms. Table 4 provides a comprehensive summary and comparison of the literature to show the potential of Metaverse simulations to guide and inform real-world decisions.

4.1 Simulated to reality (Sim2Real)

Various techniques, such as domain randomization, domain adaptation, meta-learning, and knowledge distillation, are employed to bridge the gap between simulation and reality. These techniques address the inherent mismatches during the Sim2Real transfer process, such as differences in like visual appearance, physical properties, and environmental dynamics, when models trained in simulated environments are transferred to the inherently more complex and dynamic real world.

These methods help create robust policies that can perform effectively in real-world scenarios despite the differences from the simulated training environment. Although multiple works have been done for Sim2Real in robotics (Ju et al. 2022; Salvato et al. 2021; Zhao et al. 2020), limited attention has been paid to implementing such strategies within the Metaverse. For instance, Shi et al. (2023) trained a MARL model within the Metaverse that was effectively utilized within the real world. The authors proposed recurrent multi-agent deep deterministic policy gradient (R-MADDPG) with domain randomization to bridge the simulation-to-reality gap. Domain randomization was employed to train MARL policies in a simulation environment, preparing them for real-world conditions by varying domain parameters like observation noise and environmental factors. The method is validated through simulations of multi-UAV systems in AirSim and subsequent real-world testing, demonstrating that R-MADDPG achieves superior performance and generalization. Chen et al. (2024f) address the challenge of robotic rigid contact-rich manipulation in dynamic environments focusing on the multiple peg-in-hole assembly tasks. Traditional RL methods for single peg-in-hole tasks face difficulties when applied to multiple peg-in-hole scenarios due to the increased complexity of geometric and physical constraints. To overcome these limitations, the authors leverage the industrial Metaverse to design a novel and more challenging multiple peg-in-hole assembly setup that incorporates various modalities, including vision, proprioception, and force/torque, to handle the complexity and uncertainties inherent in such tasks and improve sample efficiency. The proposed method utilizes a multimodal learning architecture that combines vision, proprioception, and force/torque into compact representations via a tokenization-based model. The policy is trained using the soft actor-critic (SAC) algorithm in a simulation environment within the Metaverse and transferred directly to real-world applications. Domain randomization and impedance control bridge the gap between simulation and reality. The proposed method comprises a pretraining approach for latent visual representations, a transformer module for multimodal dependencies, and an MLP decoder for action prediction. Experimental results demonstrate high success rates and effective generalization in simulation and real-world settings.

Liu et al. (2021) propose a framework for financial reinforcement learning in the Metaverse, FinRL-Meta, offering a data-driven of market environments. Within FinRL-Meta, the authors accelerate the training of DRL trading agents with access to large datasets and pre-processed data. The framework provides a pipeline for DRL in numerous financial trad-

Table 4 Works done to utilize the Metaverse for Reinforcement Learning models

References	Task	Proposed algorithm	Multi-agent	Simulator	Domain randomization	Replicated in real life
Shi et al. (2023)	UAV Transportation	MADDPG	✓	Airsim	✓	✓
Chen et al. (2024f)	Robotic Assembly	SAC	✗	MuJoCo, ROS, Python	✓	✓
Liu et al. (2021)	Financial Stock Trading	PPO, AC	✗	Alpaca API	✗	✓
Qin et al. (2024)	Musculoskeletal Systems Target Lifting	DDPG (feedback controller)	✗	OpenSim	✗	✗
Wang et al. (2023c)	UAV Network Trajectory Planning	DQN	✗	–	✗	✗***
Hou et al. (2024a)	Car Park Optimization	PPO HIRE-RL	✗	–	✗	✗
Ren et al. (2022)	Autonomous Vehicles	Quantaum RL	✗	Python	✗	✗
Gu et al. (2023)	Evacuation	Rainbow DQN	✗	Cinema 4D	✗	✗
Wu et al. (2023a)	Autonomous Driving	Hug-DRL*	✗	Matlab, CARLA, Pytorch	✗	✓
Wu et al. (2023b)	Autonomous Navigation	Multi-Hug**	✗	CARLA	✗	✓
Xie et al. (2024)	Human-Robot Collaboration Gear Assembly	Not Specified****	–	–	–	–
Peng et al. (2025)	Mobile robot navigation	PVP4Real	✗	MetaDrive	✓	✓
Wang et al. (2024a)	Navigation	SALM	✗	Gym-based Wang et al. (2023b)	✓	✗
Chhan et al. (2024)	Preference learning	Crowd-PrefRL	✗	DMControl suite Tassa et al. (2018)	✗	✗
Pan and Liu (2025)	Vision-Language model training	Group Relative Policy Optimization (GRPO)	✗	OpenShape	✗	✗

*Hug-DRL: human-guidance-based DRL

**Multi-Hug: Multiple Human-guided RL

***Not detailed

****The paper is proposing a framework for human-guided RL, not proposing any algorithms or reporting simulations

ing tasks and open source tools for data engineering. The PPO trading agent is trained in cryptocurrency trading and stock trading where it outperformed benchmarks in backtesting and paper trading.

Qin et al. (2024) introduces a hybrid control system for musculoskeletal robots designed to perform lifting tasks in an industrial Metaverse setting. The control system combines a brain-inspired feedforward controller, based on muscle synergies, with RL-based adaptive dynamic programming (ADP) feedback controller. The feedforward controller uses a radial basis function (RBF) network to handle phasic and tonic muscle synergies, optimizing the robot's movements for tasks like lifting. The ADP feedback controller ensures optimal control by minimizing tracking errors in real-time which was supported by an actor-critic structure to adjust muscle excitations. Experiments on an upper limb musculoskeletal system show that the hybrid controller performs well in various lifting tasks.

The RL-based model trained in the Metaverse offers significant advantages for autonomous vehicles in Metaverse applications by enhancing their adaptability and performance across virtual and physical environments. Wang et al. (2023c) introduced a novel imperfect Digital Twin (DT)-assisted RL training method to optimize multi-UAV networks while minimizing training costs. The approach leverages DT technology to simulate UAV interactions in a digital space to reduce the energy and hardware expenses associated with real-world UAV training. The authors employed two neural networks (NNs): one trained via unsupervised learning to optimize DT-related parameters and the other using RL to enhance UAV trajectory performance. Simulation results validate the proposed method by comparing convergence performance and network utility.

Multiple works utilized the Metaverse to train an agent that could be used within real-life applications. However, they limited their implementation to the Metaverse and did not perform real-life simulations. For instance, Hou et al. (2024a) presents a novel Hybrid Residual MultiExpert (HIRE) RL approach to address the complexities of high-density parking (HDP) scheduling in stack-based parking lots. Traditional HDP scheduling methods, mainly heuristic, often fail to effectively utilize information in dynamic environments. HIRE combines multiple heuristic scheduling strategies and an RL-trained neural network to optimize vehicle parking and retrieval. The framework employs an expert assignment strategy network to select the best heuristic method based on the current state and incorporates a residual network to bypass these strategies when necessary. The proposed hybrid approach enhances the RL agent's training speed, scheduling performance, and adaptability to various parking lot scenarios. Experimental results demonstrate that the HIRE RL method significantly outperforms traditional heuristic and vanilla RL methods regarding parking efficiency and adaptability. The proposed method minimizes vehicle maneuvers, manages parking space effectively, and optimizes real-time parking operations. The framework's robustness to parking lot sizes and vehicle exit time estimation accuracy further underscores its practical applicability. Similarly, Ren et al. (2022) show that traditional model training suffers from insufficient sampling due to infrequent weather conditions and slow real-time updates. Therefore, they proposed Quantum Collective Learning and Many-to-Many Matching Game-Based Scheme (MMQA) for connected autonomous vehicles (CAVs) within the Metaverse environment. MMQA operates on its own time clock system, thereby accelerating the sampling process and expanding the sample size, enhancing the accuracy and efficiency of AI models used in autonomous driving scenarios. Simulation results demonstrate the effectiveness of MMQA compared to existing methods such as Many-to-Many

Matching Strategy with Random Allocation (MMRA) and random allocation schemes. MMQA achieves higher system rewards by optimizing spectrum resource allocation and intelligently selecting vehicles based on their contribution to feature space expansion and energy efficiency. A work done by Gu et al. (2023) presents a teaching building evacuation training system based on the Metaverse, employing Rainbow DQN to dynamically guide evacuees to the safest and quickest routes based on real-time data. The simulation results indicate that this approach outperforms traditional static evacuation methods, reducing evacuation time and avoiding congestion.

4.2 Human guidance

Integrating human guidance into RL frameworks has become a transformative approach to enhance the training efficiency, safety, and adaptability of RL agents. Human guidance, whether through direct intervention or preference-based feedback, serves a critical function in overcoming the exploration-exploitation trade-off inherent in RL and significantly reduces training time, especially in complex real-world applications. By leveraging human expertise, RL systems can achieve faster learning trajectories and more reliable decision-making, particularly in environments where traditional exploration strategies may be inefficient or unsafe. Multiple works have shown that incorporating human guidance to train RL agents improves performance and significantly reduces training time and unnecessary exploration of the environment, for instance, multiple works show that incorporating human guidance to train RL agents will improve performance and significantly reduce training time and unnecessary exploration of the environment. For instance, Wu et al. (2023a) proposed a real-time human-guidance-based DRL (Hug-DRL) method for training policies in end-to-end autonomous driving. The proposed method features a novel control transfer mechanism that allows humans to intervene and correct the agent's actions in real-time during training. An improved actor-critic architecture with modified policy and value networks is developed to enable the seamless fusion of human guidance into the agent's training loop, enhancing the efficiency and performance of DRL. The Hug-DRL method was validated through experiments involving 40 human participants in various autonomous driving scenarios using a human-in-the-loop driving simulator. The experiments compared Hug-DRL with other state-of-the-art DRL approaches, demonstrating its superior training efficiency and performance under human guidance. Wu et al. (2023b) later also proposed a novel approach to enhance RL for unmanned ground vehicles (UGVs) by incorporating human intelligence into the RL framework. The authors show that it is challenging to train an RL model due to limited computing resources for navigation tasks, which results in suboptimal policies and failure to handle corner cases. To mitigate these issues, the proposed human-guided RL framework allows humans to intervene during both the training phase in simulation and the deployment phase in real-world environments. The framework includes human-guided learning objectives, prioritized human experience replay, and human intervention-based reward shaping to effectively utilize human input. The experiments show that the human-guided RL approach improves goal-reaching and safety performance compared to existing learning-based and model-based navigation strategies. The authors demonstrated the effectiveness of their model by training the RL agent in simulated- and real-world scenarios while maintaining robust performance. Additionally, the study highlights the framework's ability to fine-tune the RL agent online using small-scale human intervention, enhancing the

agent's adaptability and practical utility in diverse and dynamic environments. The results indicate that the human-guided RL framework significantly enhances the RL agent's performance, efficiency, and safety, making it a promising approach for UGV navigation tasks. Xie et al. (2024) utilized XR and industrial Metaverse to develop a human-robot collaboration assembly system. The authors utilized VR, XR, blockchain and DT technologies to improve human-robot interactions in manufacturing. The proposed system integrates physical assembly, VR-based monitoring, decision-making, and control, using a combination of virtual simulation and real-world tasks to optimize human-robot task allocation. AR is used by operators to assist with assembly and decision-making through visual aids. A T-series spiral bevel gear reducer, a complex and high-precision mechanical component, is used to show the effectiveness of the proposed system. Experimental results show that the system reduced operator workload, shortened preparation and completion times, and improved task execution. Peng et al. (2025) introduced PVP4Real, a hybrid online learning method that combines imitation learning with reinforcement learning, specifically designed for mobile robots operating in dynamic, unstructured environments. Unlike traditional RL, it enables real-time learning from human feedback that could be adapted in multiple applications. The proposed framework demonstrated that effective policies could be trained in as little as 15 min, an order of magnitude improvement over traditional methods, underscoring its potential for rapid deployment in real-world scenarios. Similarly, Wang et al. (2024a) developed the Social Robot Planner (SRLM), a hybrid framework that combines large language models (LLMs) with deep reinforcement learning to facilitate human-in-the-loop navigation in social environments. SRLM interprets real-time human language commands to inform global planning and employs a DRL-based planner for low-level motion execution. This integration allows robots to adapt dynamically to human instructions, enhancing their navigational capabilities in complex, human-populated spaces. Chhan et al. (2024) introduced Crowd-PrefRL, a framework that aggregates preference-based feedback from multiple human users to train RL agents. By estimating the reliability of each user's input, Crowd-PrefRL effectively learns reward functions that reflect collective human preferences. This approach demonstrated superior performance compared to models trained on individual or majority-vote preferences, particularly in scenarios with diverse user opinions. Metz et al. (2024) presented a comprehensive taxonomy categorizing various forms of human feedback in RL, encompassing dimensions such as feedback timing, modality, and granularity. This framework aids in designing more effective human-in-the-loop systems by aligning feedback mechanisms with specific learning objectives and agent capabilities.

4.2.1 RL from human feedback and LLMs for the Metaverse

Building on the broader theme of human guidance in RL, recent advances in reinforcement learning from human feedback (RLHF) have shown particular promise for aligning large language models (LLMs) with user preferences and values. RLHF has been widely adopted in LLM training pipelines, most notably in models like InstructGPT, where human feedback signals are used to fine-tune reward models and guide agent behavior (Ouyang et al. 2022). Unlike direct intervention or isolated preference feedback, RLHF provides a scalable framework for shaping complex model outputs through structured reward modeling, thereby aligning AI agents with nuanced human expectations. In the Metaverse, RLHF-enabled LLMs can serve as powerful mediators between humans and autonomous agents.

For instance, conversational avatars and non-playable characters (NPCs) can adapt their behaviors to user feedback, ensuring interactions remain personalized, safe, and engaging. Similarly, RLHF can guide LLM-driven decision-support systems to interpret natural language instructions, explain agent policies in accessible terms, and continuously refine their alignment with diverse user groups. These capabilities are especially relevant in shared virtual spaces, where multiple stakeholders may have conflicting preferences; RLHF frameworks that aggregate feedback across users offer pathways to collective alignment (Chhan et al. 2024). Furthermore, Pan and Liu (2025) introduced MetaSpatial, a first RL-based framework that enhances 3D spatial reasoning in vision-language models (VLMs) for scene generation in AR/VR, metaverse, and gaming. Using a multi-turn refinement strategy with three-level rewards (format, physical, and rendering-based), it progressively improves object layouts for coherence, plausibility, and realism. Experiments show large gains in format correctness, collision reduction, and GPT-4o–assessed scene quality, enabling VLMs to generate functional, high-quality 3D scenes without heavy post-processing. Moreover, more recently, Reinforcement Learning from AI Feedback (RLAIF) (Bai et al. 2022; Lee et al. 2024) is introduced as a cost-effective alternative to RLHF, using off-the-shelf LLMs to generate preference labels. RLAIF achieves similar performance to RLHF across tasks like summarization, helpful, and harmless dialogue generation. It also outperforms a supervised fine-tuned baseline, even when the AI labeler is the same model as the policy. Lee et al. (2024) introduced direct RLAIF (d-RLAIF), where a pretrained LLM directly serves as the reward function during RL training. RLAIF offers a scalable, human-free solution with comparable or better alignment to human preferences.

4.3 Simulators to realize the Metaverse

The Metaverse provides a comprehensive, interactive virtual environment for training RL agents in dynamic, persistent, and multi-agent settings. It enables the development of robust, transferable policies through large-scale, realistic interactions and human-in-the-loop feedback, minimizing the need for real-world fine-tuning. Realizing such environments and immersive experiences can be achieved using diverse simulation tools for multiple applications such as gaming, VR, and autonomous systems. These tools are compared in Table 5 and briefly categorized in this subsection.

Autonomous driving Simulators such as CARLA, AirSim/Colosseum, LGSVL, MetaDrive, and SUMO offer urban-scale driving environments with realistic traffic, pedestrian behavior, and weather dynamics. They are widely used for RL-based vehicle navigation, control, real world interatctions and interventions during training. For example, Wu et al. (2023a) employed CARLA with human guidance to train end-to-end autonomous driving agents, while Shi et al. (2023) leveraged AirSim for UAV multi-agent coordination.

Robotics Gazebo, CoppeliaSim, PyBullet, Isaac Sim, and ARGoS are widely adopted for robotic manipulation, locomotion, and embodied AI research. Open-source simulators such as Gazebo and CoppeliaSim allow rapid prototyping and integration of teleoperation interfaces, making them accessible for exploratory RL experiments. In contrast, proprietary platforms like Isaac Sim and Omniverse provide industrial-grade physics fidelity, realistic graphics, and seamless integration with digital twins. Meng et al. (2024) used Isaac Sim to test the performance of the proposed Constraint Proximal Policy Optimization (C-PPO) for synchronization tasks in the Metaverse using a real-world robotic arm.

Table 5 Simulation Tools used for Metaverse realization. The simulators are compared to highlight their accessibility and native support for RL and Human-in-the-loop integration

Simulator	Agent type	Environment	Languages	Supporting OS	Human guidance
Open source					
Apollo	Car	Urban	C++, Python	Linux	✗[1]
ARGoS	Robotics	Indoor, Outdoor	C++	Windows, Linux	✗[1]
Blender	Various	Various	Python	Windows, macOS, Linux	✗
Colosseum (AirSim)	Car, UAV	Urban, Rural	C++, Python	Windows, Linux	✓
CoppeliaSim	Robotics	Urban, Indoor	Lua, Python, Java	Windows, Linux	✓
CARLA	Car	Urban	Python, C++	Windows, Linux	✓
Gazebo	Robotics	Urban, Indoor	C++, Python	Windows, Linux	✓
Kilombo	Robotics	Swarm Robotics	C	Windows, Linux	✗
LGSVL	Car	Urban	Python, C#	Windows, Linux	✗[1]
MetaDrive	Car	Urban	Python	Windows, Linux	✓
MuJoCo	Various	Various	Python, C++, C#, C	Windows, Linux, MacOS	✗[1]
OpenSim	Neuro-musculo-skeletal				
PyBullet	Robotics	Urban	Python	Windows, Linux	✓
Sim4CV	Various	Urban	Python	Windows, Linux	✗[1]
SUMO	Traffic	Urban	C++, Python	Windows, Linux	✗[1]
Proprietary					
Isaac Sim	Robotics	Urban, Indoor	Python	Windows, Linux	✗[1]
Omniverse	Car, UAV, VR	Urban, Rural	Python	Windows, Linux	✗[1]
Simulink	Car, UAV, Robotics	Urban, Rural, Various	MATLAB, Simulink	Windows	✗[1]
Unity	Car, UAV, VR	Urban, Rural	C#, JavaScript	Windows, macOS, Linux	✓
Unreal Engine	Car, UAV, VR	Urban, Rural	C++, Blueprint	Windows, Linux	✓

[1]Possible with some engineering

Biomechanics and human modeling OpenSim enables detailed musculoskeletal and biomechanical simulations, supporting RL applications in rehabilitation, assistive devices, and human–robot collaboration. Qin et al. (2024) used OpenSim to evaluate a hybrid RL controller for robotic lifting tasks.

Immersive and social environments Unity, Unreal Engine, Blender, and Omniverse can be utilized for enhanced visual realism, multi-user interaction, and immersive experiences. These platforms facilitate rich avatars, social presence, and collaborative RL scenarios (Long et al. 2023, 2025).

5 Challenges and open issues

Despite the promising potential of RL for realizing the Metaverse, the integration of RL within the Metaverse presents a unique set of challenges. These challenges span multiple domains, including the intricacies of designing effective reward functions and the scalability and computational burden of training agents across vast and dynamic virtual environments, while ensuring that models generalize well from simulated to real-world applications. This section explores the key hurdles faced by researchers working at the intersection of RL and the Metaverse, highlighting the technical, ethical, and practical obstacles that must be addressed to unlock the full potential of an RL-empowered Metaverse (Table 6).

5.1 Training performance

Design and optimization of reward functions are key issues when developing RL-based applications for the Metaverse. In traditional RL, an agent learns to navigate an environment and make decisions based on carefully crafted reward signals. However, in the complex and dynamic environments of the Metaverse, defining success and measuring performance can be exceptionally difficult. The reward signals in such environments may often be sparse or poorly aligned with the true objectives, necessitating reward shaping (Ng et al. 1999). While reward shaping can make learning more efficient, it also introduces the risk of spurious correlations and reward hacking, where agents exploit specific aspects of the reward structure to achieve high rewards without actually fulfilling the intended tasks (Skalse et al. 2022). This misalignment between the reward and the desired outcome poses a significant obstacle in deploying RL effectively in the Metaverse (Kaufmann et al. 2023). Furthermore, RL agents often require millions of samples to learn an effective policy, making the process computationally expensive and potentially impractical, in real-life scenarios, especially, in the interactive and immersive settings of the Metaverse.

5.2 Scalability and computational complexity

Deep RL demonstrated remarkable success across various domains, including Go and Atari games. However, these successes are often limited to single-task performance, where an agent is trained exclusively on each task. A major challenge arises when these methods are simultaneously scaled to handle multiple tasks (Espeholt et al. 2018). Current state-of-the-art techniques like A3C and UNREAL require substantial computational resources, sometimes processing up to a billion frames over several days to master a single domain. Scaling such methods to tens or hundreds of tasks in parallel becomes impractical due to the immense computational burden (Espeholt et al. 2018). Although, utilizing GPUs could accelerate training such as batched A2C to processes actions and environments in parallel batches to speed up training (Clemente et al. 2017). However, this approach can be severely constrained by the slowest environment in each batch, especially when dealing with visually or physically complex simulations that require more processing time. The variance in environment speed and the initialization of episodes in variable-length environments further complicate scaling that leads to inefficiencies and bottlenecks that hinder the utilizing of RL algorithm in the Metaverse at scale.

Table 6 Comparison of current works based on Metaverse- and RL-related criteria

References	Multi-agent	Domain randomization	Replicated in real life	Human guidance	Simulator	Blockchain
Shi et al. (2023)	✓	✓	✓		✓	
Ye et al. (2024), Long et al. (2023), Hare and Tang (2023), Pu et al. (2024)	✓				✓	
Lotfi et al. (2024)	✓				✓[2]	
Kang et al. (2024a), Wang et al. (2023e)	✓					✓
Kang et al. (2024b), Zhou et al. (2023), Chen et al. (2024e), Kang et al. (2024c), Park et al. (2024b), Feng et al. (2025), Long et al. (2025)	✓					
Yu et al. (2023d), Yu et al. (2023b), Si et al. (2023a), Chen et al. (2024a), Chua et al. (2023), Chua et al. (2025), Si et al. (2023b)	✓[1]					
Chen et al. (2024f)		✓	✓		✓	
Wu et al. (2023a), Wu et al. (2023b)			✓	✓	✓	
Liu et al. (2021), Meng et al. (2024), Meng et al. (2023), Jin et al. (2022)			✓		✓	
Wang et al. (2023c)			✓[3]			
Xie et al. (2024)				✓		✓
Ren et al. (2022), Wu et al. (2023c)					✓	✓
Yu et al. (2024), Li et al. (2023b), Gu et al. (2023), Lin et al. (2022a), Hou et al. (2024b), Lin et al. (2022b), Qin et al. (2024), Wang et al. (2024c)					✓	
Zhang et al. (2023b)					✓[2]	
Yu et al. (2023c), Lin et al. (2023)						✓
Li et al. (2023c), Du et al. (2024), Chu et al. (2024), Xu et al. (2022), Park et al. (2024a), Mazandarani et al. (2023), Hou et al. (2024a), Chua et al. (2022), Kim (2024), Yu et al. (2023a), Aliyu et al. (2024), Yu et al. (2023e), Yu and Zhao (2023), Zhang et al. (2023a), Tan et al. (2023), Huang et al. (2024), Hoa et al. (2023), Gao et al. (2023), Mazandarani et al. (2025), Zhang et al. (2024), Zhang et al. (2025), Mao et al. (2024)						

[1] Hybrid-Action Agent

[2] Dataset only

[3] Unclear

5.3 Meta-to-real transfer

Meta-to-real transfer involves training algorithms within a Metaverse environment and deploying them in real-world applications. Training complex RL algorithms in the Metaverse provides the advantage of experimenting with extreme scenarios without the risks and constraints of real-world testing. It accelerates the development cycle and reduces costs compared to traditional, resource-intensive methods. However, challenges such as the sim-to-real gap can arise when transitioning from virtual to physical environments, due to the dynamic and unpredictable nature of the real-world.

5.4 Model generalization

One of the significant limitations of traditional RL models is their ability to generalize across different environments and conditions. While RL methods are designed to optimize policies based on expected cumulative rewards within a given MDP framework, they often struggle when faced with uncertainties such as random perturbations, model inaccuracies, or adversarial conditions. Furthermore, these methods rely on recurrent neural networks (RNNs) or CNN to process sequential inputs and maintain state information. However, these architectures can struggle with long-term dependencies and scaling issues (Hu et al. 2024a).

5.5 Model transparency

Currently, the RL models developed for Metaverse applications work in a black-box manner that lacks interpretability. These models make decisions based on complex calculations across multiple neural network layers; however, they are unable to justify their decisions to humans, including domain experts and even the engineers who designed the systems. This poses significant challenges when such models are deployed in critical domains, especially in healthcare or self-driving cars. Furthermore, since it is difficult to interpret how decisions are made, it becomes challenging to ensure that the models adhere to ethical guidelines and are not biased. Additionally, due to lack of interpretability, it is challenging to identify errors or suboptimal behavior in RL models.

5.6 Performance metrics

Performance metrics are indispensable for evaluating and optimizing Metaverse applications, ensuring they meet the high demands of seamless and efficient user experiences. These metrics provide essential insights into various aspects of system performance and reliability. For example, latency metrics like total latency, uplink delay, and downlink delay are vital for real-time interactions and smooth virtual experiences. Throughput and reliability assess the system's capacity to handle data traffic and maintain consistent performance under varying conditions. Visual and sensory quality metrics, such as successful frames per second (FPS), are crucial for ensuring immersive and fluid virtual environments. Furthermore, metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE) are used to evaluate the performance of ML models. Similarly, for RL models, Convergence time, reward, etc., are used to measure the agent's ability to learn and adapt to the environment. Although various Metaverse applications have unique requirements and challenges, it

is necessary to adopt a diverse set of metrics tailored to specific scenarios beyond the communications aspects of the Metaverse. There is still a need to establish common or standardized performance metrics for fair comparison and evaluation.

5.7 Security

The Metaverse, as a complex and immersive virtual environment, introduces several significant cybersecurity challenges that need to be addressed to ensure the safety and security of its users and infrastructure. The Metaverse faces multiple security challenges, which primarily include: data integrity, identity verification, and digital identity management (Awadallah et al. 2024). Data tampering and man-in-the-middle attacks can distort the digital twin's representation and lead to flawed decisions. Furthermore, the reliance on customizable digital avatars and the potential for deepfake technology to create highly realistic but fraudulent representations of users complicates verifying true identities. This leads to identity theft and impersonation, where malicious actors could deceive others, leading to unauthorized access, financial fraud, and severe privacy breaches. Additionally, integrating biometric data, such as facial features and behavioral patterns, in VR technology raises concerns about the security of sensitive personal information. If authentication mechanisms are compromised, it could result in significant privacy violations and misuse of biometric data. The dynamic nature of the Metaverse is vulnerable to advanced cyber threats like Denial of Service (DoS) attacks, which could disrupt services and cause significant downtime. Furthermore, the use of AI techniques by malicious actors to launch automated attacks that can evade detection, known as Advanced Persistent Threats (Alshamrani et al. 2019), poses a severe threat to the Metaverse's security. Traditional cybersecurity measures that rely on static approaches, such as firewalls and antivirus software, are insufficient to counter these advanced threats, particularly in the face of zero-day attacks that exploit previously unknown vulnerabilities (Ahmad et al. 2023). RL systems, which are critical in managing these complex environments, are also vulnerable to a range of security threats that can compromise the performance and integrity of the systems. RL models can be compromised through data poisoning or adversarial attacks, leading to incorrect actions that could have severe consequences, especially in critical applications like autonomous vehicles (Lei et al. 2023).

5.8 Privacy concerns

RL systems can be exploited to expose sensitive information due to the dynamic and extensive data collection involved in training. Privacy attacks may involve inferring private information from the collected data or exploiting vulnerabilities in data transfer and storage systems. Several techniques, such as differential privacy and encryption, are employed to safeguard data, while federated learning protects individual privacy by decentralizing the learning process. Most of the studies performed for Metaverse applications overlooked the security and privacy concerns associated with their proposed models. Therefore, future studies should also focus on safeguarding their proposed methods from potential security and privacy concerns.

Challenges	Research Directions		
Training Performance	RL from Human Feedback	Decentralized RL	Robust RL Models
Scalability & Computational Complexity	Decentralized RL	Robust RL Models	
Security	AI-driven Security Measures	Decentralized RL	
Model Transperancy	AI-driven Security Measures	Privacy Preserving RL	Explainable RL
Privacy Concerns	Privacy Preserving RL	AI-driven Security Measures	
Ethical Considerations	Explainable RL		
Meta-to-Real Transfer & Model Generalization	Robust RL Models	Domain Randomization in the Metaverse	
Standardized Performance Metrics	Meta Layer & Physical Layer Models	Explainable RL	

Fig. 8 The research directions for the challenges faced by RL-based Metaverse

5.9 Ethical considerations

The ethical implications of RL-driven decisions and behaviors become increasingly significant as the Metaverse becomes more immersive and interactive. Such unique ethical challenges must be addressed at multiple levels, from system design to user interaction. One of the primary ethical considerations in developing RL systems for the Metaverse is establishing a robust regulatory framework. Governments and regulatory bodies need to define clear guidelines and standards to ensure that ethical practices are embedded into the development process of RL engines, tools, and frameworks (Sami et al. 2024). For example, integrating ethical regulations directly into the design phase can prevent potential misuse of RL technologies, such as biased decision-making algorithms or unfair reward structures that might benefit certain groups of users over others. The design of RL systems in the Metaverse should adhere to ethical principles prioritizing user well-being, and ensuring that RL-driven experiences do not exploit psychological vulnerabilities, such as addiction or compulsive behavior. Developers must also obtain informed consent from users, particularly when RL systems modify or influence their behavior (Lee and Hu-Au 2021). Ensuring ethical consideration can create a Metaverse that is innovative, engaging, and aligned with ethical principles that safeguard the interests and well-being of all users.

6 Research direction

As the Metaverse continues to evolve, the integration of reinforcement learning offers vast potential but also presents significant challenges that require careful design mechanisms and innovative solutions. This section discusses possible future directions to address the key limitations posed by integrating RL techniques and applications within the Metaverse. Figure 8 shows the potential research directions for the challenges faced by RL-based Metaverse.

6.1 Reinforcement learning from human feedback

Human-in-the-loop (HITL) RL integrates human judgments, preferences, or corrections into learning. By leveraging human expertise, RL algorithms can be guided toward desired behaviors more efficiently than relying on predefined reward functions or exploration strategies, particularly in complex environments where specifying exact reward functions can be challenging. HITL RL accelerates learning and improves the alignment of AI systems with human values and expectations (Xie et al. 2024; Wu et al. 2023b). Traditional RL methods often require extensive interactions with the environment to learn effective policies, which can be computationally expensive and time-consuming. Human feedback can help bridge this gap by providing high-level guidance or corrections enabling the agent to learn more effectively from fewer interactions. However, integrating human feedback into RL algorithms can be noisy, subjective, and inconsistent, which introduces biases or errors into the learning process. Therefore, the feedback should be consistent, reliable, and accurate to reflect the desired behavior. Furthermore, the integration process must be carefully designed to balance human input with the agent's learning, preventing over-reliance on feedback that might limit the agent's ability to explore and learn independently. The dynamic nature of the Metaverse adds complexity to the collection and integration of feedback. Additionally, addressing privacy and ethical concerns related to collecting and utilizing human feedback will be essential to maintaining user trust and ensuring the responsible development of Metaverse technologies. Therefore, this presents a promising area for future research in the field.

6.2 Meta layer and physical layer

As represented in Sect. 3, the majority of existing research is focused on the Enabler layer, often neglecting other critical layers within the framework. While enablers play a foundational role in ensuring the seamless operation of Metaverse environments, the meta layer and physical layer have received comparatively little attention. For instance, Fig. 5 shows that over 80% of the works focus on the enabler layer, whereas only a small fraction address the virtual and physical layers. Therefore, future research should prioritize the unexplored layers to enhance both physical and virtual experiences and achieve seamless integration.

6.3 Decentralized RL approaches

Future research should focus on decentralized learning approaches, where the learning process is distributed across multiple agents or nodes. This method can reduce the computational burden on a single entity and improve the scalability of RL systems. Furthermore, Quantum computing can drastically accelerate RL computations, enabling faster training and more complex models. Quantum-enhanced RL algorithms could leverage quantum theory to explore and exploit large state and action spaces more efficiently than classical approaches. Moreover, employing decentralized agents enhances the security of the Metaverse by eliminating the potential risks associated with centralized systems, such as single point of failure.

6.4 Domain randomization in the Metaverse

To mitigate the issues of model generalization and Meta-to-Real transfer, techniques like domain randomization can be employed to enhance the fidelity of virtual simulations. Ensuring that the virtual environment closely mirrors real-world conditions, combined with rigorous validation processes, can significantly improve the effectiveness of meta-to-real transfer.

6.5 Robust RL models

Robust RL aims to address the challenges of Meta-to-Real transfer by extending the traditional MDP framework to robust MDPs, which incorporate additional considerations to handle environmental uncertainties. Moreover, constrained MDPs are employed to ensure safety and robustness, especially in complex applications like self-driving cars, where unobserved confounders can lead to spurious correlations that negatively impact decision-making. Interestingly, Transformers can address the challenges of RNN-based modules through their self-attention mechanism, allowing them to dynamically weigh the importance of different parts of the input sequence while remembering longer sequences. The self-attention mechanism enhances the model's ability to understand and predict the future states across various RL paradigms, including offline RL, online RL, hierarchical RL, meta-RL, and multi-agent RL (MARL) (Yuan et al. 2024). For instance, in online RL and MARL, Transformers can be utilized in a pre-train-finetune manner to leverage large-scale foundation models (Bommasani et al. 2021) for knowledge transfer to improve generalization and performance. Moreover, Transformer-based MARL approaches, such as MADT, utilize Transformers for joint policy optimization (Yuan et al. 2024). Although much work has been done to utilize transformers-based models for RL applications, these models are yet to be adapted for Metaverse applications. Therefore, there is a need to incorporate robust transformer-based models for Metaverse applications.

6.6 AI-driven security measures

The Metaverse and RL security concerns can be addressed by incorporating AI-driven cybersecurity measures that can adapt and respond to the dynamic and complex nature of the Metaverse. For instance, continuous authentication methods utilizing multi-modal biometrics—such as facial recognition, voice patterns, and behavioral biometrics—can enhance identity verification processes and prevent impersonation. These AI-based systems can detect real-time anomalies and provide more robust protection against identity theft and deepfake-related threats. Additionally, integrating blockchain technology more deeply into the Metaverse's infrastructure can enhance the security of digital transactions and asset trading. By leveraging blockchain's decentralized and transparent nature, the risk of fraud and unauthorized access to digital assets could be minimized. There is also a need to develop more resilient and adaptive defense mechanisms to counter increasingly sophisticated cyber threats. Research into AI-powered intrusion detection systems (IDS) that can learn from emerging attack patterns and proactively defend against Advanced Persistent Threats (APTs) is crucial. Moreover, enhancing the interoperability of security protocols across dif-

ferent virtual environments within the Metaverse will be essential to ensure seamless and secure interactions between users and systems.

6.7 Privacy preserving RL solutions in the Metaverse

Mitigating privacy risks in the RL-based Metaverse requires a comprehensive and multi-level approach that addresses both the inherent technical vulnerabilities and the behavioral patterns of users in these immersive environments. A privacy-by-design approach should be adopted from the beginning to ensure robust privacy protection, embedding privacy features directly into the platform's architecture and design (Wang et al. 2023a). For example, strict data minimization and anonymization can be applied to ensure that data cannot be easily traced back to individual users if data is intercepted. Advanced cryptographic techniques and secure data-sharing methods reinforce privacy in RL algorithms within the Metaverse (Lei et al. 2023). Differential Privacy (DP) can be employed to add noise to data, ensuring individual user information remains indistinguishable within larger datasets, allowing RL models to learn effectively without compromising privacy (Lei et al. 2023). Blockchain technology provides a decentralized and transparent ledger, making it particularly relevant for RL in smart contracts and decentralized applications by ensuring data integrity and security (Xiao et al. 2023). Additionally, Federated Learning enables the training of RL models on decentralized data sources, where models are trained locally on user devices, and only the trained parameters are shared with a central server. This approach preserves user privacy while allowing RL algorithms to leverage a diverse and extensive dataset.

6.8 Explainable RL solutions

Explainable Reinforcement Learning (XRL) focuses on making the decision-making processes of RL agents more transparent and explainable to enhance trust, regulatory compliance, and user interaction, especially in high-stakes applications like autonomous driving and healthcare. XRL provides insights into how agents make decisions, which can improve debugging, algorithm development, and user customization (Milani et al. 2024). Although multiple works have proposed and employed XRL methods (Vouros 2022; Milani et al. 2024) for RL model transparency and interpretability, there is still a need for more research on designing interpretable agents that offer transparent explanations of their behavior within the Metaverse. Such explanations are more meaningful than post-hoc explanations that only provide plausible justifications without true insight into the agent's decision-making process (Bekkemoen 2024). Furthermore, robust XRL models should be developed that could be generalized across various domains without a specific task or particular application.

7 Conclusion

The intersection of reinforcement learning and the Metaverse presents both exciting opportunities and significant challenges, where RL algorithms offer promising solutions for creating adaptive, intelligent systems that can enhance the user experience within the immersive virtual environments that compose the Metaverse. Therefore, this review introduces a comprehensive overview of the potentials contributions of RL to the Metaverse. It high-

lights key enabling technologies, such as, digital twins, blockchain, artificial intelligence, extended reality, and edge computing with a focus on the simulation software required to create complex virtual environments empowered by RL technology. The review studies the existing literature to understand the symbiotic relationship between RL and the Metaverse; examining 1) RL-based solutions for addressing common Metaverse problems such as scalable, efficient communications, and realistic, immersive virtual experiences and 2) Metaverse applications for providing more robust and efficient RL model training.

Despite the potential of the collaboration between RL and the Metaverse, there still exists multiple challenges, such as designing effective reward mechanisms, managing the computational complexity of training agents across vast and dynamic virtual environments, ensuring secure, scalable, and generalizable RL models that are adaptable to real-life applications, while addressing concerns related to model transparency and user privacy. In order to mitigate these challenges and realize the full potential of an RL-empowered Metaverse, possible research directions are introduced, such as, human-in-the-loop mechanisms, decentralized RL systems, privacy-preserving RL models, AI-driven security measure, and transparent and explainable RL models.

This review serves as a foundational step in advancing the mutually-beneficial relationship between the Metaverse and reinforcement learning. As the Metaverse paradigm continues to evolve, the ongoing advancement of RL techniques designed for the Metaverse, and those trained within it, will play a crucial role in shaping its future, driving innovation, and creating a more immersive, interactive, and user-centric virtual world. Thus, inviting further exploration at the intersection of reinforcement learning and the Metaverse.

Author contributions N.E., F.S., and M.S. wrote the main manuscript and prepared all figures. All authors contributed substantially to the conception and intellectual content content of the paper. All authors reviewed and revised the manuscript.

Funding The research reported in this manuscript was conducted without external funding, and the authors did not receive financial support or sponsorship from any organization that could have influenced the research outcomes.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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